NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY

A STATE UNIVERSITY

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Azad Hind Fauj Marg, Sector-3, Dwarka, New Delhi-110078

SCHEME OF COURSES AND EXAMINATION

FOR

BACHELOR OF TECHNOLOGY

BIOTECHNOLOGY

(Effective from the Session: 2019-2020)

APPROVED BY

The Senate in its II to VII meetings

The Board of Management in its meeting held on August 14, 2019

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1. INTRODUCTION

NSUT has embarked on its journey towards excellence in academics through the introduction of a novel system of learning that is being followed in many reputed universities globally. The Choice Based Credit System (CBCS) has been proposed by University Grants Commission (UGC) on recommendations of the National Knowledge Commission, to improve the quality of higher education in India. NSUT proposes to adopt CBCS for its Bachelor of Technology courses

CBCS is the mother of student centric educational reforms. A student is provided with an academically rich, highly flexible learning system blended with abundant provision for skill practice and activity orientation that he/she could learn in depth without sacrificing his/her creativity. A student can exercise the option to decide his/her own pace of learning- slow, normal or accelerated plan and sequence his/her choice of paper, learn to face challenges through term work/ project work and may venture out to acquire extra knowledge/ proficiency through add-on facilities. The great advantage of CBCS is that the learning process is made continuous and the evaluation process is not only made continuous but also made learner-centric and is designed to recognize the capability and talent of a student.

2. CURRICULUM STRUCTURE

B.Tech. programme of the University shall be based upon CBCS and shall have well defined Programme Educational Objectives (PEOs).All the courses shall have well-defined Course Outcomes (COs).Courses shall be of three kinds: Core, Elective and Foundation.

a. Core Course (CC): This is a course which is to be compulsorily studied by a student as a core requirement to complete the requirements of the B.Tech. programme.
b. Elective Course: This is a course which can be chosen from a pool of elective courses. It is intended to support the discipline of study by providing an expanded scope, enabling exposure to another discipline/domain and nurturing a student's proficiency and skill. An elective may be of the following types:

i. Discipline Centric Elective (ED): It is an elective course that adds proficiency to the students in the discipline.
ii. Generic Elective (EG): It is an elective course taken from other engineering subjects and enhances the generic proficiency and interdisciplinary perspective of students.
iii. Open Elective (EO): It is an elective course taken from a common pool of non-engineering disciplines that broadens the perspective of an engineering student. These electives shall comprise two groups: Open electives of the Humanities, Social Sciences and Management group and Open electives of the Sciences group.

c. Foundation Course: A Foundation course leads to knowledge enhancement and provides value-based training. Foundation courses may be of two kinds:
i. Compulsory Foundation (FC): It is based upon the content that leads to fundamental knowledge enhancement in Sciences, Humanities, Social Sciences and Basic engineering. They are mandatory for all disciplines.
ii. Elective Foundation (FE): It can be taken from among a common pool of foundation courses which aim at value-based education. They may provide hands-on training to improve competencies, skills or provide education on human, societal, environmental and national values. These shall be mandatory, non-credit courses, which do not carry any credits but a student has to pass in order to be eligible for award of degree.

2.1 EVALUATION AND ASSESSMENT

The performance of a student in a semester shall be evaluated through continuous class assessment, MSE and ESE. Both the MSE and ESE shall be University examinations and will be conducted as notified by the CoE of the University. The marks for continuous assessment (Sessional marks) shall be awarded at the end of the semester. The continuous assessment shall be based on class tests, assignments/tutorials, quizzes/viva-voce and attendance etc. The MSE/ESE shall comprise of written papers, practicals and viva-voce, inspection of certified course work in classes and laboratories, project work, design reports or by means of any combination of these methods.

The weightage of each of these modes of evaluation for the different types of courses shall be as per Table 1. Further, the mechanism for continuous assessment shall be as per Table 2.

Table-1: Evaluation Scheme

S. No.	Type of Course	Continuous Assessment (CA) Theory	Mid-Semester Examination (MSE) Theory	End-Semester Examination (ESE) Theory	Continuous Assessment (CA) Practical	End-Semester Examination (ES) Practical
1	FE courses	Continuous Assessment only (100 marks)				
2	CC/FC/ED/EG/EO Theory with Tutorial	25	25	50	Nil	Nil
3	CC/FC/ED/ EG/EO Theory with Practical	15	15	40	15	15
4	Project I and Project II	Nil	Nil	Nil	40	60
5	Training	Nil	Nil	Nil	40	60
6	Work shop based Course	30	--	20	30	20
7	Audit Courses*	-	-	-	-	-
*The distribution of marks of practical and/or theory components for Audit courses shall be determined by the respective Departments.						

Table 2: Continuous Assessment

S. No.	Type of Course	Continuous Assessment (CA)
1	CC/FC/ED/EG/EO Theory with Tutorial	Two class tests, Assignments, Teachers' assessment (quizzes, viva-voce, attendance)
	CC/FC/ED/EG/EO Theory with Practical	One class test, One Lab test, Assignments/Projects, Teachers' assessment
2	FE courses ***	Two class tests, Assignments, Teachers' assessment
3	Project I /II	Mid-Semester Presentation, Report, Supervisor's

		Assessment
4	Training	As specified by the Department
5	Audit Courses	As specified by the Department

*** Foundation Elective Courses are value-based courses which may enhance the proficiency /skill. These electives could be communication skills, Spoken English, soft skills, Business and Management courses, entrepreneurship development, Knowledge of an additional Foreign Language, Personality Development through sports, music, theatre, dance, etc.

The University provides to the students a pool of Foundation elective courses which may be offered by the following departments of the University:

i) Department of Humanities
ii)Department of Management
iii)Department of Personality Development

Note for Undergraduate students--

i) An Undergraduate student will have the liberty to choose any three foundation elective course to study from the given list.
ii) He/She can take only one foundation elective course in an ongoing semester.
iii) The study and clearing of foundation elective course is to be done by the end of 2nd year (fourth semester).
iv) For getting a Degree, it is mandatory to clear the entire three chosen foundation elective course.
v) Foundation elective courses are auditable course and there is no credits awarded to the students.

Note for Course Teacher--

The evaluation of the student is done through continuous assessment.

[Subject having Theory only] --The course teacher evaluate through TWO class tests (25 marks each), ONE Assignment/Project (40 marks) and internal evaluation [one such component is attendance] (10 marks).

[Subject having Theory and Practical]—Here, a course teacher evaluate for theory part through TWO class test (20 marks each) and internal evaluation [one such component is attendance] (10 marks). Similarly, for practical part ONE practical test (40 marks) and internal evaluation [one such component is attendance] (10 marks)

[Subject having Practical only]—The course teacher takes TWO practical test (45 marks each) and internal evaluation [one such component is attendance] (10 marks)

2.2SEMESTER WISE COURSE/CREDIT DISTRIBUTION

Table 3 :

Semester	Types of courses as per NSUT Nomenclature						Courses/credits	Credits
	FE (NON-CREDIT)	FC	CC	ED	ED/EG/EO/ EO- Sciences / EO-SS & Mgmt	Training Project etc.		
I	01	05	00	00	00	00	06 courses 20 credits	84 credits
II	00	03	03	00	00	00	06 courses 24 credits	
III	01	00	05	00	00	00	06 courses 20 credits	
IV	01	00	05	00	00	00	06 courses 20 credits	
V	00	00	04	--	--	00	04 –07 courses 16-28 credits	86 credits
VI	00	00	04	--	-	00	04 –07 courses 16-28 credits	
VII	00	00	00	--	--	06	00 – 05 courses 06-26 credits	
VIII	00	00	00	--	--	08	00 – 05 courses 08-28 credits	
TOTAL CREDITS								170

- **ED : At least 4 courses (16 credits)**
- **EO- Sciences : At least 1 courses (04 credits)**
- **EO-SS & Mgmt : At least 2 courses (08 credits)**
-

2.3 COURSE CODE NOMENCLATURE

The courses of various B.Tech programmes shall be assigned a course code as per the following nomenclature

2.3.1 COURSE/DEPARTMENT/SPECIALIZATION/BRANCH CODING.

The courses of various B.Tech programmes shall be assigned a course code as per the defined nomenclature (Given later). This nomenclature shall use course/department/specialization/branch coding which are defined as given below.

TABLE 4: COURSE/DEPARTMENT CODES

XX	Course Category Code	FC	Foundation Core
		FE	Foundation Elective
		EO	Open Elective
YY		BT	Bio Technology
		CH	Chemistry

	Name of Department Code	CP	Computer Engineering, East Campus
		CS	Computer Science & Engineering
		CW	Civil Engg, West Campus
		EE	Electrical Engineering
		EC	Electronics & Communication Engineering
		EP	Electronics & Communication Engineering, East Campus
		HS	Humanities
		IC	Instrumentation & Control Engineering
		IT	Information Technology
		IW	Information Technology, West Campus
		ME	Mechanical Engineering
		MG	Management
		MP	Manufacturing Process & Automation
		MT	Mathematics
		MW	Mechanical Engineering, West Campus
		PD*	Personality Development
		PH	Physics

Note: Second Alphabet P indicates East Campus, and W indicates West Campus

TABLE 5: B.TECH SPECIALIZATION/BRANCH CODES

ZZ	BT	Bio Technology
	CA	Computer Science & Engineering with Artificial Intelligence
	CB	Computer Science and Engineering (Big Data Analytics) (CSDA)(NSUT EAST CAMPUS)
	CD	Computer Science and Engi)neering (Data Science) (CSDS)
	CE	Civil Engineering (CE)(NSUT WEST CAMPUS)
	CG	Geoinformatics (GI)(NSUT WEST CAMPUS)
	CI	Computer Science and Engineering (IOT) (CIOT)(EAST CAMPUS)
	CM	Maths & Computing
	CO	Computer Science & Engineering, Main Campus
	EA	Electronics and Communication Engineering (Artificial Intelligence and Machine Learning) (ECAM) (NSUT EAST CAMPUS)
	EC	Electronics & Communication Engineering
	EI	Electronics & Communication Engineering (Internet of Things)
	EE	Electrical Engineering
	IC	Instrumentation & Control Engineering
	II	Information Technology (Internet of Things) (IIOT)(NSUT WEST CAMPUS)
	IN	Information Technology (NetwoRk security)
	IT	Information Technology
	ME	Mechanical Engineering
	MP	Manufacturing Process & Automation
	MV	Mechanical Engineering (Electric Vehicles) (MEEV)(NSUT WEST CAMPUS)

2.3.2 B.TECH COURSE CODE NOMENCLATURE

FOUNDATION CORE AND ELECTIVE COURSES AND OPEN ELECTIVE COURSES:

Course Category		Offering Department Code			Course No.	
X	X	Y	Y	0	*	*

** can take numeric values only

XX and YY may be chosen as given in Tables 1,2:

*PD offers FE courses like music, dance, yoga, sports, NSS, etc. A BOS for FE courses of PD nature (like Music, Dance, Yoga, NSS, etc), has be constituted with Dean Academics as the chairperson.

OTHER CORE AND ELECTIVE COURSES:

Program Code		Offering Department Code		Course Category	Course No.	
Z	Z	Y	Y	C/E	*	*

** can take numeric only;

C for Core and E for Elective (Discipline Centric);
YY and ZZ may be chosen as given in Tables 1,2.

2.3.3MOOCS (NPTEL BASED) FOUNDATION ELECTIVE COURSES AND OPEN ELECTIVE COURSES:

Course Category		Offering Department (NPTEL) Code		UG/PG	Course No.	
X	X	F	F	G	*	*

** can take numeric only;

XX	Course Category Code	FE	Foundation Elective
		EO	Open Elective
FF	Name of Offering Department Code for NPTEL	NH	Humanities & Social Sciences
		NM	Management
		NP	Personality Development
		NS	Sciences
G	UG/PG	0	B.Tech
		I	M.Tech

2.3.4 STUDENT ROLL NUMBER NOMENCLATURE:

Students shall be assigned roll numbers as given below.

1. B.Tech:

Year of Admission	U	ZZ (FROM TABLE 2)	4 DIGIT NUMBER

3 SEMESTER WISE COURSE ALLOCATION

3.1 COURSE ALLOCATION FOR SEMESTER I

Refer to the separate manual on Equivalent courses and scheme of courses with pre-requites if you want to find the linked courses and pre_requisites for opting for EG courses.

B.Tech BIOTECHNOLOGY (BT) - SEMESTER I														
Course Code	Type	Course	L	T	P	Credits	Evaluation Scheme (Percentage weights)					Offering Dept.	AICTE COURSE TYPE
							Theory			Practical			
							CA	MS	ES	CA	ES		
FCMT001	FC	Mathematics-I	3	1	0	4	25	25	50	-	-	Maths	BASIC SCIENCES
FCCS002/ FCHS005	FC	Computer Programming /English	3	0	2	4	15	15	40	15	15	COE/ IT	ENGG SCIENCES/ HUSS
FCEC003	FC	Electronics and Electrical Engineering	3	0	2	4	15	15	40	15	15	ECE/ ICE /EE	ENGG SCIENCES
FCPH004/ FCCH008	FC	Physics/ Environment Science and Green Chemistry	3	0	2	4	15	15	40	15	15	PHYSICS/ CHEMISTRY	BASIC SCIENCES
FCME006	FC	Basics of Mechanical Engg.	4	0	0	4	25	25	50	-	-	MPAE/ ME	ENGG SCIENCES
FEXXxxx 2*	FE	Elective Foundation	-	-	-	NIL	-	-	-	-	-	-	MANDATORY COURSE
			28 contact hours 2*			20							

1: One week induction program as per AICTE norms. Classes for I semester will commence one week later.
2*: The course codes, LTP distribution and Evaluation Scheme for Foundation Electives are given in Table 3 (list under preparation). The actual weekly load depends upon the Foundation Elective Course.

Students of the Departments of Group I shall be offered courses as follows:

1. **Semeter I : Computer Programming, Physics**
2. **Semeter II : English, Environment Science and Green Chemistry**

Students of the Departments of Group II shall be offered courses as follows:

1. **Semeter I : English, Environment Science and Green Chemistry**
2. **Semeter II : Computer Programming, Physics**

3.2 COURSE ALLOCATION FOR SEMESTER II

B.Tech BIOTECHNOLOGY (BT) - SEMESTER II														
Course No.	Type	Course	L	T	P	Credits	Evaluation Scheme (Percentage weights)					Offering Dept.	AICTE COURSE TYPE
							Theory		Practical				
							CA	MS	ES	CA	ES		
FCHS005 /FCCS002	FC	English/ Computer Programming	3	0	2	4	15	15	40	15	15	COE/IT	ENGG SCIENCES/ HUSS
FCMT007	FC	Mathematics-II	3	1	0	4	25	25	50	-	-	MATHS	BASIC SCIENCES
FCCH008/ FCPH004	FC	Environment Science and Green Chem./ Physics	3	0	2	4	15	15	40	15	15	CHEMISTRY/ PHYSICS	BASIC SCIENCES
BTPHC01	CC	Physics of Materials	3	0	2	4	15	15	40	15	15	PHYSICS	PROGRAM CORE
BTCHC02	CC	Advance Chemistry	3	0	2	4	15	15	40	15	15	CHEMISTRY	
BTBTC03	CC	Introduction to Biotechnology	3	0	1	4	25	25	50	-	-	BT	
			24 2*			24							
2*: The actual weekly load depends upon the Core Courses offered by the Department													

3.3 COURSE ALLOCATION FOR SEMESTER III

B.Tech BIOTECHNOLOGY (BT) - SEMESTER III													
Course No.	Type	Course	L	T	P	Credits	Evaluation Scheme (Percentage weights)					Offering Dept.	AICTE COURSE TYPE
							Theory			Practical			
							CA	MS	ES	CA	ES		
BTBTC04	CC	Biochemistry	3	0	2	4	15	15	40	15	15	BT	PROGRAM CORE
BTITC05	CC	Data Structures & Algorithms	3	0	2	4	15	15	40	15	15	IT	ENGG. SCIENCES
BTBTC06	CC	Cell Biology	3	0	2	4	15	15	40	15	15	BT	PROGRAM CORE
BTBTC07	CC	Chemical Engg. Principles	3	1	0	4	25	25	50	-	-	BT	ENGG. SCIENCES
BTBTC08	CC	Microbiology	3	0	2	4	15	15	40	15	15	BT	PROGRAM CORE
FE***02*	FE	Elective Foundation	-	-	-	NIL	-	-	-	-	-	-	MANDATORY COURSE
			20*			20							
*: The actual weekly load depends upon the Core Courses defined by the Department													

3.4 COURSE ALLOCATION FOR SEMESTER IV

B.Tech BIOTECHNOLOGY (BT) - SEMESTER IV													
Course No.	Type	Course	L	T	P	Credits	Evaluation Scheme (Percentage weights)					Offering Dept.	AICTE COURSE TYPE
							Theory			Practical			
							CA	MS	ES	CA	ES		
BTBTC09	CC	Structural Biology	3	0	2	4	15	15	40	15	15	BT	PROGRAM CORE
BTBTC10	CC	Bioanalytical Techniques	3	0	2	4	25	25	50	-	-	BT	PROGRAM CORE
BTBTC11	CC	Molecular Biology	3	0	2	4	15	15	40	15	15	BT	PROGRAM CORE
BTBTC12	CC	Genetics	3	1	0	4	15	15	40	15	15	BT	PROGRAM CORE
BTBTC13	CC	Immunology	3	0	2	4	15	15	40	15	15	BT	PROGRAM CORE
FE***02*	FE	Elective Foundation	-	-	-	NIL	-	-	-	-	-	-	MANDATORY COURSE
			20*			20							
*: The actual weekly load depends upon the Core Courses defined by the Department													

4. SYLLABUS OF COURSES

4.1 SYLLABUS OF FOUNDATION COMPULSORY COURSES

Course No.	Title of the Course	Course Structure	Pre-requisite
FCMT001	**Mathematics - I**	**3L - 1T - 0P**	**None**

COURSE OUTCOMES (COs):

1. Analyze and test Infinite Series and its convergence,
2. Successive differentiation and expansion of the function,
3. Curvature and Radius of Curvature in different coordinate systems,
4. Applications of definite integral,
5. Consistency of system of equations, Eigenvalue and Eigenvector.

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO11	PO12
CO 1	3	2	2	2	2	-	-	-	-	-	-	-
CO 2	3	2	2	2	2	-	-	-	-	-	-	-
CO 3	3	2	2	2	2	-	-	-	-	-	-	-
CO 4	3	2	2	2	2	-	-	-	-	-	-	-
CO 5	3	2	2	2	2	-	-	-	-	-	-	-

COURSE CONTENT:

UNIT-I
Infinite Series: Tests for convergence of series: p-series (with proof), Comparison of ratios, Ratio, Integral, Raabe's, Logarithmic and Cauchy's nth root (all tests without proofs), Alternating series, Absolute convergence, Conditional convergence. Function of Single

UNIT-II
Variable: Hyperbolic functions, inverse hyperbolic function, successive differentiation, Leibniz theorem, Taylor's and Maclaurin's theorems (without remainder terms).

UNIT-III
Curvature: Polar Curves, Differential coefficients of length of arc, Cartesian, polar and parametric forms, pedal equation, Angle between tangent and radius vector, Curvature and Radius of Curvature in Cartesian, polar and pedal forms.

UNIT-IV
Applications of definite integral: Asymptotes (in Cartesian), elementary knowledge of curve tracing, area, length, surface area and volume of revolution (in Cartesian, parametric and polar co-ordinates).

UNIT-V

Matrices: Elementary row transformation, Rank of matrix, consistency and inconsistency of system of simultaneous equations, solution of non-homogeneous and homogeneous equations, Eigenvalue and Eigenvector, Characteristic equation, Cayley-Hamilton theorem. Modal matrix

SUGGESTED READINGS:

1. G.B. Thomas, "*Calculus and Analytic Geometry"* (Pearson Education)
2. Erwin Kreyszig, "*Advanced Engineering Mathematics"* (Wiley Publication)
3. Michael Greenberg, "*Advanced Engineering Mathematics"* (Pearson Education)
4. R. K. Jain and S.R.K. Iyenger, "*Advanced Engineering Mathematics"* (Narosa Publication)
5. B. S. Grewal, "*Higher Engineering Mathematics"* (Khanna Publication)

Course No.	**Title of the Course**	**Course Structure**	**Pre-requisite**
FCCS002	**Computer Programming**	**3L - 0T - 2P**	**None**

COURSE OUTCOMES (COs):

1. To understand the basic terminology and program structures used in computer programming to solve real worldproblems.
2. To understand the need for continuing to learn new languages to solve complex problems in differentdomains.
3. To learn the process of representing problems and writing,compiling and debuggingprograms.
4. To develop programming skills in using different types of data, decision structures, loops functions, pointers, data files and dynamic memoryallocation/de-allocation.
5. To be able to code using Procedural and Object-Oriented languages.

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO11	PO12
CO 1	3	2	2	2	2	-	-	-	-	-	-	-
CO 2	3	2	2	2	2	-	-	-	-	-	-	-
CO 3	3	2	2	2	2	-	-	-	-	-	-	-
CO 4	3	2	2	2	2	-	-	-	-	-	-	-
CO 5	3	2	2	2	2	-	-	-	-	-	-	-

COURSE CONTENT:

UNIT-I

Basics of C: Basic features of C Language like Identifier, Keywords, Variable, data types,

Operators and Expression, basic screen and keyboard I/O, Control Statements, iteration, nested loops, Enumerated data types, bitwise operators, C Preprocessor statements. [6 hours]

UNIT-II

Arrays and Pointers: One and multidimensional dimensional arrays, strings arrays, operations on strings, Array and Pointers, Pointer to Pointer, other aspect of pointers, User Defined Data Types: Structures, Unions. [6 hours]

UNIT-III

Functions: Concept of modular programming, Using functions, Scope of data, Recursive functions, Pointers and functions, Command line arguments.
Files: Types of files, working with files, usage of file management functions.
[6 hours]

UNIT-IV

Overview of Object Oriented Programming: Python Programming, Concepts and Terminology. Data Types and Collection Data Types: Identifiers and keyword, Integral types floating point types, operations and formatting, Sequence types, Tuples, named Tuples, lists, set Types, sets, frozen sets, mapping types, Dictionaries, Iterating and Copying collections iterators and interactable operations and functions copying collection.

Central Structures and Functions: Conditional branching, looping, Exception handling catching and raising exceptions, custom exceptions custom functions, Names and Docstrings, Argument and Parameter unpacking, Accessing variables in Global scope, lambda functions. [9 hours]

UNIT-V

Modules and Packages: Packages, custom modules, overview of python's standard library, string handling, mathematics and Numbers, Times and dates, File formats, Data persistence.
File Handling: Writing and Reading binary data, raw binary data, compression, parsing text files, Random Access binary files, generic binary record file class.
[9 hours]
Guidelines for practical work:
Programs based on concepts of above languages.

SUGGESTED READINGS:

1. B. W. Kernighan and D.M. Ritchie, "*The C programming language*", Prentice Hall.
2. Herbert Schildt and Tata McGraw Hill, "*The Complete Reference*".

3. *O Reilly Learning Python*
4. Mark Summerfield "*Programming in Python 3: A Complete Introduction to the Python Language Pearson*"

Course Type	**Title of the Course**	**Credits**	**Course Structure**	**Pre-Requisite**
FCEC003	**ELECTRONICS AND ELECTRICAL ENGINEERING**	**4**	**3-0-2**	**None**

Course Outcomes:

1. To understand the basics of AC and DC circuits, transformers along with DC generator and motor
2. To analyze series-parallel RLC circuits and
3. To implement basic circuits using diodes, BJTs and op-amps as circuit elements
4. To get familiarized with OP-AMP and its applications
5. To develop circuits using basic electrical and electronic components

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO11	PO12
CO 1	3	2	2	2	2	-	-	-	-	-	-	-
CO 2	3	2	2	2	2	-	-	-	-	-	-	-
CO 3	3	2	2	2	2	-	-	-	-	-	-	-
CO 4	3	2	2	2	2	-	-	-	-	-	-	-
CO 5	3	2	2	2	2	-	-	-	-	-	-	-

COURSE CONTENT

Unit-I

Electric Circuits: Basic Circuit Elements, Nodal and Loop Analysis,

Superposition, Thevenin's Theorem & Norton's Theorem and Maximum Power Transfer Theorem;

Unit-II

Steady-state analysis of AC circuits: Sinusoidal and phasor representation of Voltage and current, single phase AC circuit, behavior of R, L and C

Combination of R, L and C in series and parallel, Resonance; Introduction to three-phase circuits, Star-Delta Transformation

Unit-III

Transformers: Principle of operation and construction of single-phase transformer, Introduction to DC Motor.

Electronics Devices and Circuits: Junction Diode, Applications: rectifiers, clipping and clamping circuits, LEDs;

Unit-IV

Bipolar-junction Transistor: Physical operation, operating point, load-line, Self-bias circuit, single-stage CE amplifier configuration

Ideal op-amp, inverting, non-inverting and unity gain amplifiers, integrator, differentiator, summer/subtractor.

Unit-V

Digital circuits- Boolean Algebra, logic gates, K-Maps upto 4-variables, Combinational circuits: Adders and subtractors.

Flip-Flops: SR, JK, D, T and their characteristic tables.Introduction to Sensors, Introduction to Embedded Computers.

List of experiments for Electrical and Electronics Engineering

1. Verification of Maximum Power Transfer theorem
2. Verification of Thevenin's and Norton's theorems
3. Study of resonance in series RLC and parallel RLC circuits
4. Analysis of step-up and step-down transformer
5. Implement of series RC circuit as differentiator and integrator. Also perform their analysis as low pass and high pass filters
6. Implementation of clipping and clamping circuits
7. Implementation of half-wave and full wave rectifier circuits
8. Application of LEDs in electronic circuits
9. Implementation of CE amplifying configuration. Plot gain vs frequency graph
10. Implementation of Adders and subtractors.
11. Implementation of JK and Toggle flip-flops. Subsequently implement 3-bit asynchronous up-counter.
12. Measurement of power in single phase circuits using three voltmeter and three ammeter method.
13. Experiments with common sensors
14. Experiment with embedded computers

Suggested Reading:

1. M.E. Van Valkenburg, "*Network Analysis"* Pearson Publishers 3rd Edition
2. Boylestad and Nashelsky, "*Electronic Devices and Circuit Theory*" Pearson publishers, 10th Edition
3. Edward Hughes, "*Electrical and Electronic technology*", Pearson publishers, 10th Edition
4. Malvino and Leach, "*Digital Principles and Applications*", TMH publishers, 8th Edition

Course No.	Title of the Course	Course Structure	Pre-Requisite
FCPH004	**Physics**	**3L-0T-2P**	**None**

COURSE OUTCOMESS (CO):

1. Knowing important concepts and phenomena linked to relativity
2. The concept of waves and oscillations are useful for doing analytical and numerical calculations for measurements, observations and gravitational wave communications.
3. The course is helpful to the students in understanding various optical wave phenomena which are required for optical & electromagnetic wave communications and in opticaldevices.
4. To develop the basic understanding of laser for gaining advance knowledge in the field of optical communication and opto-electronics.
5. The Concepts of Optical Fiber for modern developments in physics which are helpful in designing and developing new devices used in optical communications,medicine, environment, Industries and related physics.

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO 1	3	2	2	2	2	1	-	-	-	-	-	-
CO 2	3	2	2	2	2	1	-	-	-	-	-	-
CO 3	3	2	2	2	2	1	-	-	-	-	-	-
CO 4	3	2	2	2	2	1	-	-	-	-	-	-
CO 5	3	2	2	2	2	1	3	-	-	-	-	-

COURSE CONTENT:

UNIT-I

Relativity: Special Relativity, Lorentz Transformations, Velocity addition, Time dilation, Length Contraction, Variation of mass with velocity, Mass and energy, Relativistic momentum and relativistic energy, General theory of relativity, Einstein's theory of Gravitation, Gravitational waves, Gravity and Light.

UNIT-II

Oscillations and Waves: Damped and forced oscillations, Sharpness of resonance, Q-factor, Application in resonance, Acoustic waves, Pressure wave equations, Intensity pressure relation, Acoustic impedance, Reflection and transmission of acoustic waves, Impedance matching; Ultrasonics and its applications.

UNIT-III

Optics: Interference: Interference due to thin films, Newton's rings, and determination of the wavelength of sodium light, Interference due to wedge shaped film. Diffraction:

Fraunhofer diffraction due to single slit and N Slits, Plane transmission grating, Rayleigh criterion of resolution, Resolving power of a grating, Polarization: Polarization in light, Birefringence, Nicol prism, Quarter and half wave plates, Production and analysis of plane, Circularly and elliptically polarized light, Optical rotation, specific rotation,Polarimeter.

UNIT-IV

Lasers: Absorption and emission of radiation, Main features of a laser, Spatial and temporal coherence, Einstein Coefficients, condition for light amplification, Basic requirement for Laser, Population Inversion - Threshold Condition, Line shape function, Optical Resonators, Three level and four level systems. Classification of Lasers: Solid State Laser-Ruby laser and Gas Laser- He-Ne laser (Principle, Construction and working), Optical properties of semiconductor, Semiconductor laser (Principle, Construction and working), Applications of lasers in the field of medicine, Industry, Environment and Communication.

UNIT-V

Fibre Optics: Need for fiber Optic Communication, Physical nature of Optical fiber, Theory of Light propagation in optical fiber, Acceptance angle and numerical aperture, Step index and graded index fibers, Single mode and multimode fibers, Losses in optical fiber, Optical Fiber cables and bundles, Dispersion in optical fibers: Intermodal and Intramodal dispersion.

List of Experiments:

1. To determine the value of "g" by Bar Pendulum and find the radius of gyration.
2. To determine the wavelength of He-Ne laser.
3. To find the numerical aperture and angle of acceptance of optical fiber.
4. To find the resolving power of a telescope.
5. To find the wavelength of sodium light by Newton's ring.
6. To find the wavelength of sodium light by Biprism.
7. To find the wavelength of Mercury green line by diffraction grating using spectromet
8. To find the focal length of combination of two lenses by Nodal slide assembly and ve the formula.
9. To find the specific rotation of canesugar by polarimrter.
10. To find the dispersive power of prism material using spectrometer.

Text Books:

T1 Arthur Beiser, Shobhit Mahajan, "*Concepts of Modern Physics*" Mc-GrawHill
T2 D S Mathur, ``*Mechanics*,'' S Chand &co.
T3 N. Subramaniam and Brij Lal, ``*A Text Book of Optics*,'' S Chand&Co.
T4 A K Jha "*A Text Book of Applied Physics, Volume-1*" I.K. International Publishing House.
T5 Indu Prakash, ``*A Text Book of Practical Physics, Volume-1*,'' Kitab MahalPublication.

Reference:

R1 Serwey, Moses, Moyer, ``*Modern Physics*,'' Cengage Learning

R2 Jenkins and White, ``*Fundamentals of Optics*,'' McGrawHill
R3 Ajay Ghatak "*Optics*" McGrawHill

REVISED SYLLABUS OF CORE ENGLISH

Course No.	**Title of the Course**	**Course Structure**	**Prerequisite**
FCHS 005	**Core English**	**3L 0T 2P**	**None**

Course Outcomes

CO 1: Acquire competence in Basic English grammar. Grammatical accuracy, avoiding inappropriacy and using language naturally and confidently

CO 2: Improve in the four integral skills of language and to be able to use language as a tool for effective communication

CO 3: Enable the learner to express and be understood by others with clarity and precision, in both written and spoken forms, minimizing ambiguity and verbosity.

CO 4: Understand creative use of language through translation, articles and paragraph writing.

CO 5: Reading: Encouraging the habit of reading for different purposes and to analyse, paraphrase and read critically.

CO 6: Develop competence in formal Standard English pronunciation and usage

CO 7: Build confidence to use a standard spoken form of English to face jobinterviews, and workplace interactive situations besides enabling the learner to pursue advancedprofessional courses.

COs.	Theory	Hours	Lab
1.	1. Vocabulary Enhancement CO 1 1.1 Using a standard dictionary- word spellings, meanings, usage, pronunciation, making sentences 1.2 Word collocations 1.3 Commonly misused words, verbal reasoning 1.4 One word substitutions	4	Lab Activity No 1: Phonetics: Sounds Used in English Language CO 6 Lab Activity No 2: Reading from newspapers/magazines/blogs to build up a repertoire of words CO 5

	1.5 Abbreviations & foreign phrases		
2.	2. Remedial & Applied Grammar CO1 & CO 2 2.1 Tenses & Voice 2.2 Subject-Verb Agreement 2.3Narration,Interrogative structures and Question tags 2.4 Prepositions, Pronouns and Adverbs 2.5 Redundancy 2.6 Idiomatic use of language 2.7 Identification of errors and editing	6	Lab Activity No 3: Introducing Oneself: Breaking the Ice CO 5 Lab Activity No 4: Situational & Spontaneous English(tense, registers) through Role Play CO 7 Lab Activity No 5: Question Formation & Mock Press Conference CO 5
3.	3 Techniques of Good Writing CO 5 & CO 2 & 3 3.1Writing self assessment tasks 3.2 Precis writing and note-making. 3.3 Paragraph and Essay writing. 3.4 Article writing and summarizing	10	Lab Activity No 6: Blog Writing/Creating a Newsletter Lab Activity No 7: Script writing & enacting for a street play CO 6

4.	4. Business Communication: CO 4 & CO 3 4.1 Formal and Informal Letter writing 4.2 Statement of Purpose 4.3 Job application & CV (summary statement of academic & professional profiles) 4.4Power point presentations through relevant slides.	10	Lab Activity No 7: Communication at Workplace. Develop negotiating skills by using appropriate language of courtesy Lab Activity No 8: Recording individual efforts and holding paired interactions and Group Discussions Lab Activity No 9: Preparing and practising for Interviews. CO 7
5.	5.Written Comprehension CO 3 & 4 5.1 The ability to write after listening to and reading select speeches, news bulletins, presentations and answering questions based on what has been heard. 5.2 Reading the given texts to skim, scan, infer and answer comprehension questions. 5.3 Reading texts like case studies and project reports for critical assessment. 5.4 Book Review	10	Lab Activity No 10:)Introduction to Podcast and Understanding Audio texts Lab Activity No 11: Declamation and/or speeches Lab Activity No 12: Reading, GD and presentation based on listed texts CO 7

Course No	Title of the Course	Course Structure	Pre-Requisite
FCME006	**Basics of Mechanical Engineering**	**L-T-P: 4-0-0**	**None**

COURSE OUTCOMES (COs)

After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills and attitudes:

1. To know force, its nature and applications.
2. To know the basic principles of civil and mechanical structures.
3. To understand the fundamentals of thermodynamics and fluid mechanics.
4. To know the working principles of IC Engines.
5. To understand the importance of different engineering materials.

6. To understand the different manufacturing processes and machining operations.
7. To know the use of Automation in manufacturing.

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO11	PO12
CO 1	3	2	2	1	1	-	-	-	-	-	-	-
CO 2	3	2	2	1	1	-	-	-	-	-	-	-
CO 3	3	2	2	2	2	1	-	-	-	-	-	-
CO 4	3	2	2	2	2	1	-	-	-	-	-	-
CO 5	3	2	2	2	2	1	-	-	-	-	-	-
CO 6	3	2	2	2	2	1	-	-	-	-	-	-
CO 7	3	2	2	2	2	1	-	-	-	-	-	-

COURSE CONTENT

Group A

Unit-I

Introduction to Engineering Mechanics: Rigid and Elastic bodies, Force and its type, Law of parallelogram of forces, Triangle law of forces, Polygon law of forces, Lami's theorem, Laws of motion, Moment, Couple, Varignon's theorem, Conditions of equilibrium, Concept of free body diagram, Coulomb's friction, Plane trusses, Analysis of trusses, Numerical problems. (6 Hours)

Unit-II

Introduction to Strength of Materials: Simple stresses and strains, Direct, shear, and volumetric stresses and strains, Hooke's law, Tension test, Elastic constants, Poisson's ratio, Factor of safety, Introduction to beam, Types of beams, Types of loads, Shear force and bending moment diagrams (SFD and BMD) for Simple and Cantilever beams under various loading conditions, Numerical problems. (6 Hours)

Unit-III

Introduction to Manufacturing Engineering: Classification and use of engineering materials, Basic principles and applications of methods of manufacturing such as casting, forming

and joining; Working principles and applications of machining operations such as Turning, Thread cutting, Milling, Shaping, Grinding, etc., Use of automation in manufacturing. (6 Hours)

Group B

Unit-IV

Introduction to Thermodynamics: Thermodynamic system, Cycle, Path, Thermodynamic properties, Extensive and intensive properties, Thermodynamic equilibrium, Reversible and irreversible processes, isochoric, Isothermal, Isobaric, Isentropic and Polytropic processes, First law of thermodynamics applied to a cycle and process, Kelvin-Planck and Clausius statements of Second law of thermodynamics, Carnot cycle, Entropy, Clausius inequality, Internal combustion (IC) engines, IC engines terminology, Spark ignition (SI) and Compression ignition (CI) engines, Two and four stroke engines, Air standard cycles such as Otto, Diesel, Dual and Brayton cycles, Numerical problems. (12 Hours)

Unit-V

Introduction to Fluid Mechanics: Properties of a fluid, Density, Specific volume, Specific weight, Specific gravity, Kinetic and Kinematic viscosity, Pascal's law and its applications, Laminar and turbulent flow, Use of continuity equation and Bernoulli's equation, Numerical problems. (6 Hours)

SUGGESTED READINGS

1. Beer and Johnston, "*Engineering Mechanics*" Pearson
2. D.K. Singh, "*Strength of Materials*" CRC Press
3. Nag, "*Engineering Thermodynamics*" McGraw-Hill
4. Cengel, "*Fluid Mechanics*" McGraw-Hill
5. D.K. Singh, "*Fundamentals of Manufacturing Engineering*" CRC Press

Course No.	Title of the Course	Course Structure	Pre-Requisite
FCMT007	**Mathematics II**	**3L-1T-0P**	**None**

COURSE OUTCOMES (CO)

1. **Ordinary Differential Equations,**
2. **Partial Derivatives, Maxima and Minima for functions of two or more variables,**
3. **Evaluation of double and triple integral,**
4. **Concept of Numerical Methods and its Applications,**
5. **Concept of Probability and Statistics and its Applications.**

COURSE CONTENT:

UNIT-IOrdinary Differential Equations:

Second & higher order linear differential equation with constant coefficients, general solution of homogenous and non-homogenous equations, Euler-Cauchy equation, Series solution by Frobenius method.

UNIT-IIFunction of Several Variables:
Partial Derivatives, Euler's Theorem, Total differentiations, Change of Variables, Jacobian and its basic properties, Taylor's theorem, Maxima and Minima for functions of two or more variables, Lagrange's method of undetermined multipliers.

UNIT-IIIMultiple Integrals:

Evaluation of double integral (in Cartesian and polar co-ordinates), change of order of integration, change of variables. Triple integral (in Cartesian) and its applications. Gamma and beta function.

UNIT-IVNumerical Methods:
Solution of system of linear equations using Gauss elimination method, LU decomposition method Gauss Seidel iteration method, Solution of polynomial and Transcendental equations by Newton-Raphson method, NumericalIntegration by trapezoidal rule and Simpson's 1/3 and 3/8 rule, NumericalSolutions of first order ordinary differential equations: Euler's method, Runge-Kutta method of fourth order.

UNIT-VProbability and Statistics:
Conditional probability, Random Variables, Probability distributionfunctions-binomial, Poisson, exponential, uniform and normal distributions; Correlation, rank correlation and regression analysis; Sampling Theorem.

Recommended Books:

1. G.B. Thomas "*Calculus and Analytic Geometry*" (Pearson Education)
2. Erwin Kreyszig "*Advanced Engineering Mathematics*" (Wiley Publication)
3. Michael Greenberg "*Advanced Engineering Mathematics*" (Pearson Education)
4. R. K. Jain and S.R.K. Iyenger "*Advanced Engineering Mathematics*" (Narosa Publication)
5. B. S. Grewal "*Higher Engineering Mathematics*" (Khanna Publication)
6. Anthony J. Hayter "*Probability and Statistics for Engineers*" (Cengage Learning)
7. M. K. Jain, S. R. K. Iyenger and R. K. Jain "*Numerical Methods for Scientific and Engg. Computations*" (Wiley Eastern Ltd.)

4.2 FOUNDATION ELECTIVE COURSES

4.2.1 LIST OF FOUNDATION ELECTIVES

Table 3: FOUNDATION ELECTIVES

Code	Name of Foundation Elective	L T P Allocation			Evaluation Scheme Theory Practical					Pre-Requisites
		L	T	P	CA	MS	ES	CA	ES	
FEPD001	Sports-I	0	0	4	-	-	-	100	-	None
FEPD002	Sports-II	0	0	4	-	-	-	100	-	FE001
FEPD003	NSS	0	0	4	-	-	-	100	-	None

FEPD004	NCC	0	0	4	-	-	-	100	-	None
FEMG005	Corporate Social Responsibility	2	0	0	100	-	-	-	-	None
FEPD006	Music	0	0	4	-	-	-	100	-	None
FEHS007	Basic of social sciences	2	0	0	100	-	-	-	-	None
FEHS008	Spoken Skills in English	0	0	4	-	-	-	100	-	None
FEMG009	Financial Literacy	2	0	0	100	-	-	-	-	None
FEHS010	Introduction to Ethics	2	0	0	100	-	-	-	-	None
FEHS011	Stress Management	1	0	2	50	-	-	50	-	None
FEHS012	Organizational Behavior	2	0	0	100	-	-	-	-	None
FEPD013	Theatre	0	0	4	-	-	-	100	-	None
FEPD014	Dance	0	0	4	-	-	-	100	-	None
FEPD015	Yoga	0	0	4	-	-	-	100	-	None
FEPD016	Digital Film Making	0	0	4	-	-	-	100	-	None
FEPD017	Workshop (Electrical and Mechanical)	0	0	4	-	-	-	100	-	None
FEHS018	Ethical Decision Making	2	0	0	100	-	-	-	-	None

4.2.2 SYLLABI OF FOUNDATION ELECTIVES

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD001	Sports-I	0L-0T-4P	None
COURSE OUTCOMES (CO): To evolve a higher education system that is suitability blended with provision for knowledge values and skill practice where every student learns in without sacrificing his/her creativity.			

COURSE CONTENT: (Any 2 out Of 5 Components)

A. INTRODUCTION TO PHYSICAL EDUCATION IN THE CONTEMPORARY CONTEXT (Any Two)
1. Learn and demonstrate the technique of Suryanamaskar.
2. Develop Physical Fitness through Calisthenics / Aerobics / Circuit-Training / Weight-Training and demonstrate the chosen activity.
3. Select any one game available in the college and learn different techniques involved in its play

B. CORE PHYSICAL EDUCATION-: FITNESS, WELLNESS AND NUTRITION (Any Two)
1. Measurement of Fitness Components – Leg-raise for Minimal Strength (Muscular Strength); Sit-ups Muscular Endurance); Harvard Step Test, Run and Walk Test (Cardiovascular Endurance); Sit and Reach Test (Flexibility)
2. Measuring height, weight, waist circumference and hip circumference, Calculation of BMI (Body Mass Index) and Waist-Hip Ratio
3. Engage in at least one wellness programme and write a report on it.

C. CORE PHYSICAL EDUCATION-: POSTURE, ATHLETIC CARE AND FIRST AID (Any Two)
1. Demonstrate Stretching and Strengthening Exercises for Kyphosis, Scoliosis, Lordosis, Knock Knees, Bow Legs, Flat Foot, Back Pain and Neck Pain
2. Illustration and Demonstration of Active and Passive Exercises
3. Asanas with Therapeutic Value (Any five asanas): Karnapeedasana, Padmasana, Dhanurasana, Sarvangasana, Paschimottanasana, Chakrasana, Halasana, Matsyasana, Ardhmatsyendrasana, Usthrasana, Mayurasana, Shirshasana, Vajrasana.
4. Practice P.R.I.C.E. in First Aid.

D. SPORTS ADMINISTRATION & MANAGEMENT (Any Two)
1. Demonstration of Supervision activities in Sports Management.
2. Demonstration of skills of Management.
3. Demonstration of fixtures of various kinds in sports competitions.
4. Demonstration of technical and non-technical purchase procedure.
E. Adventure Sports and Leadership Training

SUGGESTED READINGS:
1. Graham, G., "*Teaching Children Physical Education: Becoming a Master Teacher. Human Kinetics*" Champaign, Illinois, USA.
2. Corbin, C. B., G. J. Welk, W. R Corbin, K. A. Welk, "*Concepts of Physical Fitness: Active Lifestyle for Wellness"* McGraw Hill, New York, USA.
3. Anspaugh, D.J., G. Ezell and K.N. Goodman, "*Teaching Today Health,*" Mosby Publishers
4. Beotra, Alka, "*Drug Education Handbook on Drug Abuse in Sports*" Applied Nutrition Sciences, Mumbai.
5. Ammon,R., Southall , R.M. and Blair, D.A., "*Sports Facility Management, ''West Virginia*, USA: Fitness Information Technology Publishers

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD002	Sports-II	0L-0T-4P	FE001
COURSE OUTCOMES (CO): To evolve a higher education system that is suitability blended with provision for knowledge values and skill practice where every student learns in without sacrificing his/her creativity.			

COURSE CONTENT:
(Any 3 out Of 5 Components)
A. Sports for all (Any Two)
1. To participate in any intramural Tournaments (one team game and one Individual Game) of choice.
2. To participate/ attend at least 15 hours in Fitness training at Field or at Gymnasium.
3. Participate in at least one track and one field event on Annual Sports day.
4. To participate in Inter College Tournament

B. Skill Enhancement Courses (any one out of three)
1. Wellness and Fitness
2. Holistic personality Development
3. Sports Journalism
4. Mass demonstration Activities

C. MEDIA AND CAREERS IN PHYSICAL EDUCATION (Any Two)
1. Organize an event / intramural / tournament in your college.
2. Prepare a News Report of an observed Sports competition.
3. Create a presentation on any topic from Physical Education using an audio-visual aid.
4. Demonstrate Warming-up / Conditioning / Cooling-down exercises.

D. MANAGEMENT OF AEROBICS & GROUP TRAINING (Any Two)
1. Measurement of Fitness Components – Leg-raise for Minimal Strength (Muscular Strength); Sit-ups (Muscular Endurance); Harvard Step Test or Run and Walk Test (Cardiovascular Endurance); Sit and Reach Test (Flexibility)
2. Measurement of Pulse Rate / Heart Rate at Radial Artery and Carotid Artery, Calculation of Target Heart Rate
3. Developing a 5-10 minute routine of aerobics with appropriate music for each component of health related physical fitness

E. SPORTS INDUSTRY & MARKETING (Any Two)
1. Identify an issue or a trend in the sports industry: o Players in professional or college sports o Ownership
2. Marketing Plan: Environmental Factors and Product Plan Draft, Paper bibliography/works cited.
3. Sponsorship proposal
4. Developing a budget plan for an event
5. Athlete branding

SUGGESTED READINGS:
1. Covey, S. , "*7 Habits of Highly Effective People*" Covey Publications, USA
2. Magill, R.A., "*Motor Learning and Control: Concepts and Applications"* McGraw Hill Publication.
3. Masteralexis, L.P., C. Barr and M. Humms, "*Principles and Practices of Sport Management*" Jones and Bartlett Publisher
4. Bishop, J.G., "*Fitness through Aerobics*" Benjamin Cummings USA.
5. Brown K.M., "*Physical Activity and Health: An Interactive Approach*" Jones and Bartlett Publisher
6. Cornwell. T.B, "*Sponsorship in marketing: Effective communications through sports, arts and events*" Routledge Publishers
7. DeGarris, L., "*Sports Marketing: A Practical Approach*" Routledge Publishers, USA

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD003	**National Service Scheme (NSS)**	**0L-0T-4P**	**None**

COURSE OUTCOMES (CO):

1. Develop among them a sense of social and civic responsibility;
2. Utilize their knowledge in finding practical solution to individual and community problems;
3. Identify the needs and problems of the community and involve them in problem solving process;
4. Utilize their knowledge in finding practical solution to individual and community problems;
5. **Develop capacity to meet emergencies and natural disasters**

COURSE CONTENT:

Unit-I

Introduction to NSS: Orientation and structure of NSS, History of Social Reforms in Modern India: Brahmo Samaj, Arya Samaj, Satya Shodhak Samaj: Principles and Functions

Unit-II

Regular activities: Distribution of working hours- association between issues and programs- community project- urban rural activities, association- modes of activity evaluation

Unit-III

concept of society- development of Indian society: Features- Division of labors and cast system in India; Features of Indian constitution; Provisions related to social integrity and development

Unit – IV

N.S.S. Regular Activities

A) College campus activities
B) N.S.S.activities in Urban and Rural areas
C) Role of Non-Government Organisation (NGO) in social Reforms
i) Red Cross
ii) Rotary

SUGGESTED READINGS:

1. *National Service Scheme Manual*, Govt. of India
2. *Training Programme on National Programme scheme*, TISS.
3. *Orientation Courses for N.S.S. programme officers*, TISS.
4. Ram Ahuja, "*Social Problems in India*" Rawat Publication.
5. Ed. J. Y. Bhosale "*History of Social Reforms in Maharashtra*" S. U. Kolhapur

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD004	**National Cadet Corps (NCC)**	**0L-0T-4P**	**None**

COURSE OUTCOMES (CO):

1. Develop among them a sense of social and civic responsibility;
2. Utilize their knowledge in finding practical solution to individual and community problems;
3. Identify the needs and problems of the community and involve them in problem solving process;
4. Utilize their knowledge in finding practical solution to individual and community problems;
5. **Develop capacity to meet emergencies and natural disasters;**

COURSE CONTENT:

UNIT I:

Introduction to NCC, National Integration & Awareness: Religions, Culture, Traditions and Customs of India, National Integration: Importance and Necessity, Freedom Struggle.

UNIT II:

Adventure Training: – Obstacle course, Slithering, Trekking, Cycling, Rock Climbing, Para Sailing, gliding, Scuba Diving- methods and use.

UNIT III:

Environment Awareness and Conservation: Natural Resources – Conservation and Management. Water Conservation and Rainwater Harvesting

UNIT IV:

Personality Development and Leadership: Introduction to Personality Development, Factors Influencing /Shaping Personality: Physical, Social, Physiological, Philosophical and Psychological, Self-Awareness Know yourself/ Insight, Change Your Mind Set, Communication Skills: Group Discussion / Lecturettes (Public Speaking), Leadership Traits, Types of Leadership

SUGGESTED READINGS:

1. Bhogle Anita & Bhogle Harsha, "*The Winning way, Learning from sports for managers'' Westland Publications*
2. Sharma Robin, "*The leader had no title*" Simon and Schuster Ltd.

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEMG005	**Corporate social responsibilities**	**2L-0T-0P**	**None**

COURSE OUTCOMES (CO):

1. The course will help students to understand corporate and emerging social responsibility for the corporate in reference to India and global situation
2. The course will support students to prepare themselves to work with corporate understanding collective aspiration of the society, individual and corporate social responsibility.

COURSE CONTENT:

UNIT I: Corporate social responsibility in Indian context and International: CSR – Definition, concepts, Approaches of CSR, overview of corporate social responsibility and corporate social accountability, SR Tools, National and International CSR activities, corporate philanthropy, drivers of CSR, difference between corporate governance, corporate philanthropy and CSR

UNIT II: Business ethics and corporate social responsibility: Concept of business ethics – meaning, Importance and factors influencing business ethics. Corporate Governance – meaning, significance, principles and dimensions. Ethical decision – making in different culture, consumer protection, environment protection, gender issues in multiculturalism, ethics and corruption, ethics and safety. Business benefits of CSR

UNIT III: Legislative measures of CSR: Corporate, labor, stake holders, Environmental and pollution. Social Accounting, Social Auditing, SA: 8000 and Corporate Social Reporting.

SUGGESTED READINGS:

1. Harsh Srivastava, "*The business of social responsibility*" books for change
2. CV. Baxi and Ajit Prasad "*Corporate social responsibility – concepts and cases*" Excel Books
3. Dr. M. Mahmoudi, "*Global strategic management*," Deep & Deep Publications Pvt. Ltd.
4. S K. Bhatia, "*International Human resource management – Global perspective*" Deep & Deep Publications Pvt. Ltd.
5. J.P. Sharma, "*Governance, Ethics and Social responsibility of business*" Ane books Ltd.
6. Kotler Philip and Lee Nancy "*Corporate social responsibility; doing the most good for your company*" John Wiley
7. Simpson, Justine and Taylor, John R, "*Corporate Governance Ethics and and CSR*" Kogan Page Publishers

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD006	**Music**	**0L-0T-4P**	**None**

COURSE OUTCOMES (CO):

The student will be familiarized with the basic terms used in Indian classical music. Also it familiarizes with the life history of some dignitaries in the field of music. This course also throws some light on the ancient music and its origins in India.

COURSE CONTENT:

Unit 1: Study of the following terms: - Mela (Thāt), ĀshrayRāga, Rāga, Lakshana, Shruti, Alankar, Gamak, Vadi-SamvādiAnuvādi-Vivādi, VakraSwara, Varjit-Swara.

Unit 2: Biographies & contributions of the following: - Jaidev, MansinghTomar, Abdul Karim Khan, Tyagaraja, Pt. Bhatkhande, Pt. Ravi Shankar

Unit 3: Study of following Rāgas&TālaRāga- Yaman, Jaunpuri, Khamaj. Tāla- Ektāl, Jhaptāl

Unit 4: Genara**l** discussion and definition of the following: -

a. Khyāl, MaseetKhani – Razakhani gat, Dhrupad, Tarana, Meend, Soot, Murki, Kan, Khatka, Krintan, Harmony, Melody.

b. Writing of Bhatkhande Swarlipi Paddhati.

c. Writing of Tālasand Compositions in Notation.

d. Detailed study of Rāgas (Rāga- Bihag, Malkauns, Vrindavani Sarang) and comparative study of Rāgas.

e. Essay, Shastriya Sangeet (Classical Music) & SugamSangeet(Light Music)

Unit 5: Vedic Music – Samvedic Sangeet, Swara, Vadya, Bhakti, Vikār. General study of Natyashastra, SangeetRatnakar.

SUGGESTED READINGS:

1. Vasant and Laxmi Narayan Garg, "*Sangeet Visharad*" Sangeet Karyalay
2. Sarat Chandra Pranjpayee and Chowbhamda, "*BhartiyaSangeetkaItihas*," Surbharti Prakashan
3. Bharat Muni, "*NatyaShastra*,"
4. Sharangdeva, "*SangeetRatnakar*,"
5. Sharad Chandra Pranjpayee, "*Sangeet Bodh*,"
6. Thakur Jaidev Singh, "*Indian Music*," Sangeet research academy
7. V. N. Bhatkhande, "*Mallika Part II & III*," KramikPustak.
8. V. N. Patwardhan, "*RaagVigyan*,"
9. RaginiTrivedi, "*Ragvibodha Mishrabani, Vol. I & II*,"

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEHS007	**Basics of Social Sciences**	**2L-0T-0P**	**None**

COURSE OUTCOMES (CO):

Social science is a major category of academic disciplines, concerned with society and the relationships among individuals within a society. It in turn has many branches, each of which is considered a "social science".

COURSE CONTENT:

Unit I: Economics, political science, human geography, demography and sociology.

Unit II: Humanities, anthropology, archaeology, jurisprudence, psychology, history, and linguistic.

Unit III: Political science, economics, sociology, international politics and scientific methodology.

SUGGESTED READINGS:

1. A.C. Kapoor, "*Principles of Political Science*," S. Chand Publications
2. A.K. Sharma, "*Issues in Social Demography*," Mittal Publications
3. Kathy S. Stolley, "*The Basics of Sociology*," Greenwood Press.
4. Paul M. Muchinsky ,"*Psychology Applied to Work*," Thomson Learning Inc

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEHS008	**Spoken Skills in English**	**0L-0T-4P**	**None**

COURSE OUTCOMES (CO):

1. This course will focus on oral & presentation skills of students with practice sessions in the language lab.
2. This course will develop confidence building in oral skills of learners.
3. It will seek to encourage the day to day conversations/dialogues and communicative needs of learners with ample practice in the lab.
4. The theory class will boost practice in ample language exercises to encourage oral skills.
5. This will also involve practice sessions in interview skills, group discussions & pair work.
6. Basics of communication process, Barriers to Oral Communication
7. Elevator pitches - Practicals

COURSE CONTENT:

- Practice on listening and reading comprehension
- Language lab practice for group discussion and interviews
- Definition and discussion on communication & the barriers in communication with practical training to use language as a tool for sharing, discussing, handling and convincing others.

SUGGESTED READINGS:

"Everyday English I & II" Cambridge University Press/Foundation books

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEMG009	Financial Literacy	2L-0T-0P	None

COURSE OUTCOMES (CO):

1. To provide in-depth knowledge of the banking and Principles of Investment, financial planning.
2. Help students in understanding stocks, sell strategy, mutual fund options, investing in education, planning for the future, purchasing your first home, taxes and tax planning, life insurance options, health insurance, property insurance, estate planning, and keeping money in perspective.

COURSE CONTENT:

UNIT I: Banking- Definition, Role of Bank in growth of saving and Investment, Types of banks , Services offered by banks, Deposits and Loans, Types of A/c, Opening a bank A/c, How to Transact with banks, KYC norms, (A/c opening form, Address Proof), How to read bank statement, Banking products and services, Calculating Interests – Saving, FD, Simple and Compound Interest, Power of compounding Loans, Types of loans, taking a home loan, Definition of EMI, Calculation of EMI, Post office-Account and transactions, Basic of foreign Exchange, Importance and Use of Foreign Exchange, Regulator Role of RBI, mutual funds.

UNIT II: Investment: Principles of Investment – Safety, Liquidity and Return, Investment plans, Hybrid plans-Ulip, SIP and VIP of mutual funds, index funds

UNIT III: Financial Planning- Meaning, Household financial health checkup, Important life stages, Medical and other Emergencies, ; Insurance, Meaning, Need and Wants, Loss protection, Life, non-life and health, Benefits of Insurance, Term plans, Social obligations Budgeting, Buying a house, Plan a vacation, Retirement planning, Price of procrastination, Market and financial instruments, Primary market, Secondary market, Financial Statement analysis,

UNIT IV: Scams, Fraud Schemes-Insider trading, Money laundering; Consumer protection and redressal mechanism, Rights of Consumers, Applicable to financial services, Filing a complaint, Complain to entity concerned, Regulators, Arbitration, Consumer courts, Govt. Websites-(PG Portals), Investor Associations, Taxes, Meaning, Need of Taxes, Types of taxes, How taxes impact income, Income, wealth and gift tax, Service tax, STT, Stamp Duty, Tax planning v/s tax evasion, Tax rates, Tax free bonds, Tax saving investment

SUGGESTED READINGS:

1. Braunstein, Sandra, and Carolyn Welch, "*Financial literacy: An overview of practice, research, and policy*," Fed. Res. Bull.
2. Cole, Shawn A., and Gauri Kartini Shastry, "*Smart money: The effect of education, cognitive ability, and financial literacy on financial market participation*" Harvard Business School, 2009.
3. *Study material of NSE.*
4. Gitman, joehnk and Billingsley, "*Personal financial planning*," Cengage Learning
5. Madura Jeff, "*Personal finance student edition*," Prentice Hall PTR**.**

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEHS010	**Introduction to Ethics**	**2L-0T-0P**	**None**

COURSE OBJECTIVES (CO):

1. It is aimed to comprehend right from wrong, to act upon something tricky with a deliberative analysis.

Course Outcomes:

2. Helps in addressing issues with a moral reasoning and analysis.

COURSE CONTENT:

1. Fundamentals of Ethics
2. Issues in Moral Philosophy
3. Theories of Justice and their Applications
4. Ethical Decision Making
5. Applied Ethics
6. Media Ethics
7. Environmental Ethics
8. Technology & Ethics
9. Feminism

SUGGESTED READINGS:

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEHS011	**Stress Management**	**1L-0T-2P**	**None**

COURSE OBJECTIVES (CO):

The objective of this course to help the students to understand the nature of stress, sources of stress and to identify the symptoms of stress through first unit. Second Unit aims to teach the students to learn certain skills and the strategies required for effectively managing the stress and ability to cope up from the stressful situations.

COURSE OUTCOMES (CO):

1. To understand the nature, sources of stress and consequences of stress
2. To overcome from the constraints in managing stress
3. To develop the motivation to adopt different technology

COURSE CONTENT:

UNIT I: Stress (GAS Model), Learning about sources of stress and its symptoms: Nature of stress- various sources of stress environmental, social (including social media), physiological and psychological; Symptoms of stress - emotional response, physiological & behavioral; relationship between stress and performance, relationship between stress and health

UNIT II: Learning to manage stress effectively: Methods - yoga, meditation, Vipassana,

relaxation techniques, clarifying problem, alternate actions, support (Problem focused) emotion focused constructive approach, Indian Case Studies

Practical: (50 marks)
Any two practical's based on Unit I and II

SUGGESTED READINGS:

1. DiMatteo, M.R. & Martin, L.R.(2002) "*Health psychology*" New Delhi: Pearson
2. Neiten, W. & Lloyd, M.A (2007). "*Psychology applied to Modern life*" Thomson Detmar Learning .
3. Taylor, S.E. (2006). "*Health psychology*" 6th Edition. New Delhi: Tata McGraw Hill

Course No.	**Title of the Course**	**Course Structure**	**Pre-Requisite**
FEHS012	**Organizational Behavior**	**2L-0T-0P**	**None**

COURSE OUTCOMES (CO):

1. The course aims at providing a comprehensive understanding of organization (structure and culture) and its functioning, at the levels of the individual, group and organization as a whole.
2. To acquaint the students with employee motivation and work attitudes and its relationship with performance and productivity.
3. To help students gain insights into the concept of organization change in the context of ever changing business environment and provide them with various tools of organizational development.
4. To provide students an overview of organizational dynamics in the light of power, politics and stress.

COURSE CONTENT:

Unit 1: Understanding Organizational Behavior: Defining organization and Organizational Behavior (OB), OB Model; the Organizational structure; Common Organizational Designs; New Design Options

Unit 2: Employee Attitudes and Motivation: Job Satisfaction; Organizational Commitment; Organizational Citizenship behavior; Positive Organizational Behavior (POB):
Theories of Work motivation: Content theory (Maslow, Herzberg), Process theory (Vroom's Expectancy Theory, Equity Theory), Contemporary theories (Goal Setting theory and Self-Regulation theory)

Unit 3: Dynamics of Organizational Behavior: Organizational culture; Power and Politics: Influence, sexual harassment, organizational politics, Causes & Consequences of stress at the workplace:

Unit 4: Organizational change and development: concept of organizational change, model of organizational change (one model), organizational development: concepts, models (one model), techniques of organizational development

Unit 5: Organizational behavior in startups

SUGGESTED READINGS:

1 Aamodt, M. G. (2016). "*Industrial/Organizational psychology: An applied approach*" Boston: Cengage Learning.
2 Luthans, F. (2011), "*Organizational behaviour:An evidence based approach*" 12th Edition. McGraw Hill
3 Muchinsky, P.(2007), "*Psychology applied to work: An introduction to Industrial and Organizational Psychology*" NC: Hypergraphic Press.
4 Pareek, U. &Khanna,S.(2012,Third edition) "*Understanding Organizational Behaviour*" Oxford: Oxford University Press.
5 Pareek, U., & Gupta, R. K. (2010), "*Organizational behaviou"* New Delhi: Tata McGraw Hill.
6 Robbins, S.P., Judge T.A. and Sanghi, S.(2009), "*Essentials of Organizational Behaviour*" 10th edition. Pearson Education , India.
7 Schultz, D & Schultz, S (2013), "*Psychology and Work Today*" Pearson Education, India.

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD013	**Theatre**	**0L-0T-4P**	**None**

COURSE OUTCOMES (CO):

Our goal is to nurture artist-scholars who are well read in dramatic literature, who understand the social and historical contexts of that literature, who appreciate contemporary performance and dance, who think critically, who master discipline-specific skills, and who make compelling artistic choices on stage.

COURSE CONTENT:

Unit 1: Concept of Acting in Indian Classical theatre. Western styles of theatre acting.

Unit 2: Basics of the following: Acting in Grotowski's Poor Theatre, Folk Theatre of India

Unit 3: Acting for Camera –Knowledge of camera frames and movement within the confines of a frame, blocking, difference between theatre and Camera acting, Concentration.

Unit 4: Acting consistently for different takes, acting scenes out of order, Auditions, acting exercises. Art of Dubbing.

SUGGESTED READINGS:

1. Boleslavsky, Richard, "*Acting: the First Six Lessons*" New York Theatre Arts.
2. Hagen, Uta, "*Respect for Acting*" Macmillan Press.
3. Hodge, Alison, "*Twentieth Century Actor Training*" London and New York.

4. Routledge ,Stanislavski, Konstantin, "*An Actor's Work: A Student's Diary*" Trans. and ed. Jean
5. Jeremiah Comey, "*The Art of Film Acting*" Focal Press .
6. Philips B Zarrilli, "*Acting (Re) Considered*" Routeledge .
7. Cathy Hassey, "*Acting for Film*," Allworth Press
8. Singh. Y, "*Indian Sociology social conditioning and emerging concerns*," Vistaar publication.

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD014	**Dance**	**0L-0T-4P**	**None**

COURSE OUTCOMES (CO):

This course will provide the student with the fundamentals necessary for advanced dance skills. Further, this course will develop student appreciation of dance as an art form and lifetime activity. Designed to familiarize students with technique, the student will also study vocabulary, different forms of dance, issues in dance and the history pertaining to the world of dance. The student will develop kinesthetic awareness, movement memory, creative abilities and aesthetic appreciation of various dance forms. The enhancement and the development and maintenance of physical fitness, self-confidence, self-discipline and independence with the body by providing informal showings during class are the goals expected to be achieved. Each student should leave this class having been encouraged, esteemed, and take with them a new appreciation of dance.

COURSE CONTENT:

- Basic workout
- Introduction to Hip Hop and B-Boying with a simple choreography
- Exercise like: Rolling, jumping, moving shoulders. Footwork, Floor steps, Beat knowledge.
- Freestyle combination along with House dance style.
- Expressions class: Body expressions, Face expressions.
- Introduction of Contemporary Dance. Basic exercise of Contemporary Dance. Exercise for flexibility, Floor steps, Spinning and Balancing.
- Introduction to Jazz. Basic exercise and proper routine practice.

SUGGESTED READINGS:

1. Jonathan Burrows, "*A Choreographer's Handbook*," Routledge
2. Jacqueline M. Smith-Autard, "*Dance Composition: A Practical Guide to Creative Success in Dance Making*," Routledge

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD015	**Yoga**	**0L-0T-4P**	**None**

COURSE OUTCOMES (CO):

Students will learn about the importance of yoga in their lives. They will be exposed various types of yoga, their health benefits.

COURSE CONTENT:

UNIT-I

Origin of Yoga & its brief development, Meaning of Yoga & its importance, Yoga as a Science of Art (Yoga Philosophy),Meaning of meditation and its types and principles.

UNIT- II

Classification of Yoga/Types of Yoga, Hatha Yoga , Raja Yoga, Laya Yoga, Bhakti Yoga, Gyan Yoga, Karma Yoga, Asthang Yoga.

UNIT –III

Principles of Yogic Practices, Meaning of Asana, its types and principles, Meaning of Pranayama, its types and principles, Meaning of Kriya its types and principles.

UNIT -IV

Yogic therapies and modern concept of Yoga, Naturopathy, Hydrotherapy, Electrotherapy, Messotherapy, Acupressure, acupuncture, Meaning and importance of prayer, Psychology of mantras, Different mudras during prayers

SUGGESTED READINGS:

1. William Broad, "*The Science of Yoga: The Risks and the Rewards*," Simon and Schuster
2. Swami Vishnu Devananda, "*The Complete Illustrated Book of Yoga*," Harmony

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD016	**Digital Film Making**	**0L-0T-4P**	**None**

COURSE OUTCOMES (CO):

Students will learn about various technicalities involved in digital film making. They will also expose to history of cinema, preproduction etc.

COURSE CONTENT:

Unit 1 – History of Cinema, Research & Script

Early Cinema, Development of Classical Indian & Hollywood Cinema, History of Global Film including European Film (1930-present), Origin of Classical narrative cinema-Soundless film, Exploration of film and analysis of the three-part beginning, middle and end of story, **Research**(Finding and Collecting materials and facts related to your story. Where and How to find the materials related to your story. Things to consider before sketching down your story), **Script (Scriptwriting Process and its various phases), Film Grammar for Scriptwriting.**

Unit 2 – Pre-Production

Digital Video Cinematography: Introduction to Digital Video Cinematography
Cinematography, Interactivity and emotions through Cinematography,
Building blocks, Compositions, Lenses and Cameras, Types of lenses: Zoom Lens, Prime Lens, Types of Cameras: HD Cameras, Basics of Film Camera, Difference between, Film Camera and Digital Camera, DSLR and HDSLR Cameras, Lighting, Psychology of light, Visual Environment, Directional Effect of Light, Lighting design process, Three-point lighting, High-Key lighting, Low Key lighting, Construction of a Shot, Color, Contrast, Deep Focus, Shallow Focus, Depth of Filed, Exposure, Racking focus, Frame Rate, Telephoto shot, Zoom shot.

Unit 3- Digital Video Editing

Effective Editing, Principles of Video Editing, Non-Linear Editing (NLE) Concept, The Three-Point Edit, Non-Linear Editing (NLE) Techniques, Working in the Timeline, Transitions, Key framing, Applying Filters, Ingesting.

Unit-4Advanced Editing Techniques

NLE Compositing, Color Correction & Color Grading, Working on Audio, Titling

SUGGESTED READINGS:

1. Mark Brindle and Chris Jones, "*TheDigital Filmmaking Handbook*," Quercus

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD017	**Workshop (Electrical and Mechanical)**	**2L-0T-0P**	**None**

COURSE OUTCOMES (CO):

1. Student will be able to make various joints in the given object with the available work material.
2. The students will be able to understand various wiring connections

COURSE CONTENT:

Mechanical Workshop Experiments

1. Blacksmith
2. Carpentry
3. Fitting
4. Foundry
5. Welding

Electrical workshop Experiments

1. Study & Performance Of Different Types Of Wire Joints
2. Study And Performance Of Staircase Wiring
3. Study And Performance Of Series And Parallel Connection Of Flourescent Tube Light
4. Study And Performance Of Godown Wiring
5. Series And Parallel Connection Of Bulbs And Power Sockets By Single

Switch And Multi Switches.

SUGGESTED READINGS:

1. Hajra Choudhury, Hazra Choudhary and Nirjhar Roy, "*Elements of Workshop Technology, vol. I*" Media promoters and Publishers Pvt. Ltd.
2. W A J Chapman, Workshop Technology, "*Part -1, 1st South Asian Edition,*" Viva Book Pvt Ltd.
3. P.N. Rao, "*Manufacturing Technology, Vol.1*," Tata McGraw Hill
4. Kaushish J.P., "*Manufacturing Processes*," Prentice Hall

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEHS018	**Ethical Decision Making**	**2L-0T-0P**	**None**

COURSE OUTCOMES (CO):

COURSE CONTENT:

UNIT I: ETHICAL CONCEPTS AND ETHICAL APPROACHES

1. Values, Dilemma and Choices
2. Responsibility, Justice & Fairness
3. Respect for self and others

UNIT II: ETHICAL DECISION PROCESS

1. Ethical codes and tests
2. Steps to ethical decision-making
3. Case studies and Situational role plays

SUGGESTED READINGS:

- Blanchard, K., & Peale, N.V. (1988) "*The Power of Ethical Management,*" New York: William Morrow and Co. pp. 20-24. http://www.blanchardbowleslibrary.com/books/powerofethicalmanagement.htm
- Brown, M. (1996) "*The Quest for Moral Foundations: An Introduction to Ethics*" Georgetown University Press
- Davis, M. (1999) "*Ethics and The University,"* New York: Routledge.
- Heller, R. (1998) "*Making Decisions*," New York: DK.
- Josephson, M. S. (2002) "*Making Ethical Decisions*," Josephson Institute of Ethics.
- Kardasz, F. (2008) "*Ethics Training For Law Enforcement: Practices and Trends*," VDM, Verlag Dr. M.ller.
- Nosich, G. M. (2002) "*Learning to Think Things Through: A Guide to Critical Thinking*," Prentice Hall.

4.3 PROGRAM CORE COURSES

4.3.1 LIST OF PROGRAM CORE COURSES

SEM.	Code	Name of Core Course	L T P Allocation			Evaluation Scheme Theory and Practical					Syllabus page Nos
			L	T	P	CA	MS	ES	CA (P)	ES (P)	
II	BTPHC01	Physics of Materials	3	0	2	15	15	40	15	15	45
	BTCHC02	Advance Chemistry	3	0	2	15	15	40	15	15	46
	BTBTC03	Introduction to Biotechnology	3	0	1	25	25	50	-	-	47
III	BTBTC04	Biochemistry	3	0	2	15	15	40	15	15	49
	BTITC05	Data Structures & Algorithms	3	0	2	15	15	40	15	15	50
	BTBTC06	Cell Biology	3	0	2	15	15	40	15	15	51
	BTBTC07	Chemical Engg. Principles	3	1	0	25	25	50	-	-	52
	BTBTC08	Microbiology	3	0	2	15	15	40	15	15	53
IV	BTBTC09	Structural Biology	3	0	2	15	15	40	15	15	55
	BTBTC10	Bioanalytical Techniques	3	1	0	25	25	50	-	-	56
	BTBTC11	Molecular Biology	3	0	2	15	15	40	15	15	57
	BTBTC12	Genetics	3	0	2	15	15	40	15	15	58
	BTBTC13	Immunology	3	0	2	15	15	40	15	15	60

4.3.2 SYLLABI OF PROGRAM CORE COURSES : II SEMESTER

Course No.	Title of the Course	Course Structure	Pre-requisite
BTPHC01	Physics of Material	3L - 0T - 2P	None

COURSE OUTCOMES (COs):

After completion of the course, the students will be able to:

1. Understand the crystal structure and the types of bonding in materials.
2. Apply the principles of quantum mechanics for determining the properties of microscopic particles.
3. Analyze the behavior of the materials as conductors, semiconductors and insulators.
4. Assess various properties and characteristics of dielectric, magnetic and super conducting materials.
5. Design innovative electronic devices, energy storage devices, superconducting and compact-design devices based on the principles of nano technology.

COURSE CONTENT:

MODULE-I

Crystal Structures, Imperfections and Bonding in Solids - Bravais lattice, Simple crystal structures, packing fraction, Crystal plane and Miller indices, Inter planar separation, X-ray diffraction, Bragg's law, Determination of crystal structure: Laue technique, Rotating crystal method, Powder method, brief analysis of X-ray diffraction pattern.

Crystal Defects, Point defects, Schottky and Frenkel defect, Edge and Screw dislocations, Berger's vector, Effect of man-made defects on the properties of the materials.

Bonding in solids, Different kinds of bondings, Bond energy associated with simple crystal.

MODULE-II

Fundamental of Quantum Mechanics: Matter Waves and de-Broglie Hypothesis, Uncertainty Principle, Wave Packets, Interpretation of Wave function, Schroedinger Wave Equation, Simple Eigen value Problems and degeneracy, Particle in a box.

Conduction in Metals: Free electron model, Energy distribution of electrons in a metal, Fermi Dirac Probability function, Fermi energy level, Density of states, Drift velocity, Mobility of electrons, Sommerfeld theory.

MODULE-III

Introduction to Band Theory of Solids and Semiconductors: Formation of energy bands, Comparison of metals, semiconductors and insulators on the basis of energy band, Carrier concentration in intrinsic and extrinsic semiconductors, Effect of temperature and impurity on fermi level, life time, recombination process, Hall effect, Conduction and diffusion current.

MODULE-IV

Dielectric Materials: Dielectric polarization, types of polarization, local electric field, Clausius-Mossotti relation, Debye's equation and molecular structure, Dielectric breakdown, Introduction to piezoelectricity and ferroelectricity

Magnetic Materials and Superconductors: Ferromagnetism, Antiferro, Ferri-ferro magnetism, ferrites, Domain theory, Types of superconductors, Meissner effect, high temperature superconductors, future applications.

MODULE-V

Introduction to Nanoscience and Nanotechnology: Basic principles, Quantum confinement, Effects on properties at nanoscale, Preparation of nano-materials, 1D, 2D and 3D nanomaterials, carbon nanotubes–structure, properties and uses, applications of nanotechnology.

PRACTICAL:

1. Current sensitivity and resistance of a Ballistic Galvanometer
2. Sparking potential of a Neon Lamp

3. Resistivity of a semiconductor using Four Probe Method
4. Band Gap of a given specimen
5. High resistance by Leakage Method
6. Work function by using Richardson Equation
7. Susceptibility of $MnCl_2$
8. To determine the Planck's Constant,
9. Characteristics of a GM Tube, Diode characteristics
10. Solar cell characteristics, e/m by Thomson's method, Hall effect and Hall coefficient.

SUGGESTED READINGS:

Text Books:

T1 A. Bieser, "*Concepts of Modern Physics*," McGrawHill

T2 John Allison, "*Electronic Engineering Materials and Devices*," Tata McGrawHill

T3 A.S. Vasudeva, "*Modern Engineering Physics*," S. Chand Publication.

T4 A.K. Jha "*Applied Physics, Volume-II*" I.K. International Publishing House.

T5 S.O. Pillai "*Solid State of Physics*" New age Publisher.

T6 S.M. Lindsay "*Introduction to Nanoscience*" Oxford

References:

R1 A. J Dekker, "*Electronic Engineering Materials*," PrenticeHall

R2 LH Van Vlack, "*Elements of Material Science and Engg*," Addision Wesley

R3 C Kittel, "*Solid State Physics*," Wiley Publication

R4 C.P Poole and F .J Owens, "*Introduction to nanotechnology*," WileyPublication

Course No.	Title of the Course	Course Structure	Pre-requisite
BTCHC02	**Advance Chemistry**	**3L - 0T - 2P**	**None**

COURSE OUTCOMES (COs):

1. To understand and remember organic reaction mechanism
2. To know about tautomerism, pericyclic reactions etc and analyze and apply its diverse roles in Biotechnology.
3. To get an idea of non-benzenoid aromatic system and apply this in biotechnology
4. To conversant with hands-on knowledge on organic analysis
5. To provide an insight into latest (R&D oriented) topics, to enable the engineering student upgrade the existing technologies and pursue further research

COURSE CONTENT:

MODULE-I

Reaction mechanisms – Nucleophilic and electrophilic aliphatic and aromatic substitution reactions, elimination reactions, inductive effect, electromeric effect, mesomeric effect, hyperconjugation

MODULE-II

Reaction Intermediates - Carbocations, carbanions, biradicals, enamines, carbenes, nitrenes, benzyne and their reactions, the energetics of organic reactions.

MODULE-III

Alicyclic compounds – classification, synthesis of alicyclic compounds, physical and chemical properties, Baeyer's strain theory, strainless or puckered theory, the relative stability of chair and boat form, the relative stability of cycloalkanes in terms of M.O. theory, medium ring compounds, large ring compounds, natural compounds with large rings (civetone and muscone)

MODULE-IV

Pericyclic reactions - Mechanism and stereochemistry of pericyclic reactions, types of pericyclic reactions (cycloadditions, electrocyclic reactions, sigmatropic rearrangements), (4+2) cycloadditions,

(2+2) cycloadditions, electrocyclic reactions, Woodward Hoffmann rule compounds and cyclic compounds.

MODULE-V

Conformational analysis –Conformations, conformations of ethane, substituted ethanes, cyclohexane, substituted cyclohexanes, cyclohexene, cyclohexanone, fused ring compounds, other ring systems, conformations and chemical reactivity of acyclic

Non-benzenoid aromatic system-Ferrocene, fulvene, tropylium cation, tropone and tropolone, azulene, annulenes, heterocyclic compounds, sydnones.

PRACTICAL:

1. Qualitative organic analysis - Analysis of binary organic mixtures- Tests with the mixture, Separation of the mixture,
2. Purification of the compounds, Identification of components.
3. Detection of Elements (Test for Sulphur, nitrogen halogens)
4. Determination of melting and boiling points, Specific or confirmatory tests, Preparation of suitable derivatives.
5. Detection of functional groups of a given sample
6 Quantitative organic analysis - Estimation of Sulphur, nitrogen, halogens, aniline, glycine, phenol, glucose, determination of molecular weight.
7. Chemical hydrolysis of a protein sample and characterization of the products.
8. Chemical hydrolysis of a Carbohydrate sample and characterization of the products.
9. Chemical hydrolysis of a Lipid sample and characterization of the products and characterization of the products.

SUGGESTED READINGS:

1. A. Zlatkis,"*A Concise Introduction to Organic Chemistry*",McGraw Hill.
2. P. Sykes,"*A Guide Book to Mechanism in Organic Chemistry*", Longmans.
3. J.March,"*Advanced Organic Chemistry-Reactions, Mechanisms and Structure*",McGraw Hill.
4. D.E. Applequist, C.H. Depuy and K.L.Rinehart, Jr.,"*Introduction to Organic Chemistry by Publisher*", John Wiley and Sons Inc.
5. F.A. Carey and R.J. Sundberg,"*Organic Chemistry*", Plenum Publishing Corporation.
6. J. B. Hendrickson, S.H. Pine, D.J. Cram and G.S. Hammond,"*Organic Chemistry*", McGraw Hill.
7. R.T. Morrison and R.N. Boyd, "*Organic Chemistry*", Prentice Hall College Division.
8. B. Miller,"*Organic Chemistry: The Basis of Life*", Addison-Wesley Publishing Company.
9. F.G. Mann and B.C. Saunders,"*Practical Organic Chemistry*", Longman.
10. A.I. Vogel, B.S. Furniss, and A.R. Tatchell,"*Vogel's Textbook of Practical Organic Chemistry*", Addison-Wesley Publishing Company.

E Learning

1. http://www.e-learning.chemie.fu-berlin.de/en/index.html
2. https://ocw.mit.edu/courses/chemistry/

Course No.	Title of the Course	Course Structure	Pre-requisite
BTBTC03	Introduction to Biotechnology	3L - 1T - 0P	None

COURSE OUTCOMES (COs):

1. To define an idea of biotechnology from historical practices as well as day to day activities
2. To be able to understand about the highly sophisticated nature of science and engineering involved in designing a living system
3. To discuss chemistry of water, interactions in biological system and properties of biomolecules
4. To be able to apply immense potential of biological information for generating innovative products and services for varied applications

5. Students will be able to evaluate and analyse applications of Biotechnology and its significance for environment and human welfare.
6. Students will be able to formulate the concept of Biotechnology as an integrated discipline.

COURSE CONTENT:

MODULE I

History and living system History and Scope: Historical perspectives, Biotechnology as an integrated discipline, Cell as unit of life, organization and function.

MODULE II

Biomolecules and Biomolecular interactions. Chemical foundations of Biology – Properties of water,covalent bonds, Non-covalent interactions in biological systems. Carbohydrates and their functional significance – Sugars, Polysaccharides, Glycoproteins.
Lipids – classification, structure and function. Lipids and biological membranes

MODULE III

Proteins and Nucleic acids. Introduction to proteins: Amino acids and peptides – classification, chemical reactions and physical properties. Introduction to protein structure and function, Vitamins and Coenzymes,Nucleic acids – nitrogenous bases, nucleotides, types, structure and properties of nucleic acids. DNA, RNA

MODULE IV

Understanding Biological Information. Nature of biological information, its storage, maintenance and propagation, engineering of biological information

MODULE V

Applications and Ethical issues. Biotechnology–historical perspectives, Biotechnology as an integrated discipline, Biotechnology applications in Industry, food, agriculture and healthcare. Ethical and regulatory issues

SUGGESTED READINGS:

1. Walter and York "*Agriscience Fundamentals and Applications*" Delmar Publishers.
2. E.S. Grace "*Biotechnology Unzipped*" Joseph Henry Press/ National Academy Press.
3. S.R.Barnum "*Biotechnology: An Introduction*" Wadsworth Publishing Company.
4. L. Lehninger, D. L. Nelson, and M. M. Cox, "*Principles of Biochemistry*", Worth Publishing.
5. K. E. Van Holde and W. C. Johnson "*Principles of Physical Biochemistry*", Prentice Hall.
6. T. G. Cooper, "*Tools of Biochemistry*", John Wiley and Sons Inc.
7. L. Stryer, "*Biochemistry*", W.H. Freeman and Company.
8. C. K. Mathews, K. E. Van Holde and K.G. Ahern, "*Biochemistry"*, Benjamin/Cummings.
9. K. Robert, M. D. Murray, D. K. Granner, P. A. Mayes and V. I. Rodwell, "*Harper's*
   a. *Biochemistry*", McGraw-Hill/Appleton and Lange.
10. H. Segal, "*Biochemical Calculations*", John Wiley and Sons.
11. Work and Work, "*Laboratory Techniques in Biochemistry and Molecular Biology*",Elsevier Science.
12. MORELAND B. "*Lippincott's Illustrated Reviews: Biochemistry*"
13. Adams RL, "*The biochemistry of the nucleic acids*" Springer Science & Business Media; 2012 Dec 6.

Digital Resources:

1)Introduction to Biomolecules - https://bio.libretexts.org/Courses/University_of_California_Davis/
2) Protein Structure and Function - https://www.ncbi.nlm.nih.gov/books/NBK21177/
3)Applications of Biotechnology to Fermented Food - https://www.ncbi.nlm.nih.gov/books/NBK234679/
4)Genetically Engineered Crops: Experiences and Prospect - https://www.ncbi.nlm.nih.gov/books/NBK424533/
5)Emerging Trends and Products of Biotechnology - https://www.ncbi.nlm.nih.gov/books/NBK442203/

4.3.3 SYLLABI OF PROGRAM CORE COURSES : III SEMESTER

Course No.	Title of the Course	Course Structure	Pre-requisite
BTBTC04	**Biochemistry**	**3L - 0T - 2P**	**None**

COURSE OUTCOMES (COs):

1. The students will be able to define the laws of thermodynamics in biological system
2. The students will develop an understanding of metabolism
3. Students will apply the concepts of energetics to understand Mitochondria's function and Electron transport chain.
4. The students will identify and analyze how energy transformation and transactions takes place in biological systems.
5. The students will be able understand the integration and regulation of metabolism and to formulate the causes of metabolic disorders, on the basis of catabolic and anabolic pathways.

COURSE CONTENT:

MODULE I

Metabolism and bioenergetics - First and second law, free energy and chemical equilibrium, Organic reaction mechanisms, Design of metabolism - concept of ΔGO, ATP-
ADP cycle. Cellular energy transactions – role of mitochondria and chloroplast, Basics of Enzyme

MODULE II

Carbohydrate metabolism glycolysis pathway and reactions, Glycogen breakdown and synthesis, control of glycogen metabolism. Citric acid cycle, Gluconeogenesis, The glyoxylate pathway, Pentose phosphate pathway

MODULE III

Mitochondrion and electron transport chain, Proton Motive Force, ATP Synthase and production, Regulation of ATP production, Coupling and uncoupling of ETC and phosphorylation, Inhibitors of ETC, Photophosphorylation.

MODULE IV

Lipid, Nucleic acid and Amino acid metabolism –Lipid Metabolism, Nucleic acid metabolism
Amino-acids metabolism.

MODULE V

Integration and regulation of metabolism, Metabolic disorders, Recent advances in Biochemistry

PRACTICALS:

1. Preparation of solutions based on molarity, normality, percentage, dilutions etc.
2. Determination of pI of a given protein
3. Qualitative and Quantitative tests for carbohydrates
4. Qualitative and Quantitative tests for Proteins
5. Estimation of Nitrogen content by kjedahl method
6. To determine the Iodine number of the given oil sample.
7. To find pKa value of given amino acid
8. Estimation of DNA and RNA concentration

SUGGESTED READINGS:

1. A. L. Lehninger, D. L. Nelson, and M. M. Cox, "*Principles of Biochemistry*", Worth Publishing.
2. K. E. Van Holde and W. C. Johnson "*Principles of Physical Biochemistry*", Prentice Hall.
3. T. G. Cooper, "*Tools of Biochemistry*", John Wiley and Sons Inc.
4. L. Stryer, "*Biochemistry*", W.H. Freeman and Company.
5. C. K. Mathews, K. E. Van Holde and K.G. Ahern, "*Biochemistry*", Benjamin/Cummings.
6. K. Robert, M. D. Murray, D. K. Granner, P. A. Mayes and V. I. Rodwell, "*Harper's Biochemistry*",

McGraw-Hill/Appleton and Lange.
7. K. Wilson and K. H. Goulding, "*A Biologist's Guide to Principles and Techniques of Practical Biochemistry*", Print India.
8. I. H. Segal, "*Biochemical Calculations*", John Wiley and Sons.
9. Devlin's "*Textbook of Biochemistry with Clinical correlations*," John Wiley and SonsInc.
10. Lieberman M, Marks AD. Marks' "*basic medical biochemistry: a clinical approach*" Lippincott Williams & Wilkins; 2009.

Digital Resources:

1) Biochemistry, Electron Transport Chain- https://www.ncbi.nlm.nih.gov/books/NBK526105/
2) Metabolism Basic Concepts and Design- https://www.ncbi.nlm.nih.gov/books/NBK21170/

Course No.	Title of the Course	Course Structure	Pre-requisite
BTITC05	**Data Structures & Algorithms**	**3L - 0T - 2P**	**None**

COURSE OUTCOMES (COs):

1. Understand and remember algorithms and its analysis procedure.
2. Illustrate the concept of data structures through ADT including List, Stack, Queues.
3. To design, implement and compare various data structure algorithms.
4. To select appropriate data structure for representation of the data in the real world.
5. To formulate new solutions or improve existing applications using data structure algorithms.

COURSE CONTENT:

MODULE I

Introduction to data structure: Arrays, Stacks and Queues: Fundamentals and Representations, Applications of Arrays, Stacks and Queues, Sparse Matrices, Space and time complexity of algorithms.

MODULE II

Linked lists: Singly/Linear Linked lists, Linked list implementation of Stacks and Queues, Doubly and Circular Linked Lists, Applications.

MODULE III

Trees: Binary Trees, B-Trees, N-ary Trees, B+Trees, Tree Traversals, infix to post fix conversion, construction of tree from preorder and inorder notation, Storage of Trees, Threaded trees, AVL trees, Red and Black trees, Application of Trees, Hashing, Heaps: Types of heaps, minheap, maxheap, double heap.

MODULE IV

Graphs: Types, Terminology and Representations, Graph Traversals, Applications of Graphs.

MODULE V

Searching and Sorting: Sequential and Binary Searching, Search trees, Sorting Techniques.

SUGGESTED READINGS:

1. 1. E. Horowitz and S. Sahani, "*Fundamentals of Data Structures in C",* 2nd Edition, Universities Press.
2. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein "*Introduction to Algorithms*", 3rd Edition, The MIT Press Cambridge, Massachusetts London, England McGraw-Hill Book Company.
3. Mark Allen Weiss, "*Data Structures and Algorithm Analysis in C",* Addison-Wesley.
4. Schaum's Outline Series, "*Data Structure*", TMH, Special Indian Ed., Seventeenth Reprint.
5. Y. Langsam et. al., "*Data Structures using C and C++"*,PHI
6. N. Dale and S.C. Lilly, "*Data Structures*" D.C. Heath and Co.

7. R. S. Salaria, Khanna, "*Data Structure & Algorithms*", Book Publishing Co. (P) Ltd..
8. Richard F. Gilberg and Behrouz A. Forouzan, "*Data Structure A Pseudocode Approach with C*", Cengag Learning, 2nd Ed.,

Course No.	Title of the Course	Course Structure	Pre-requisite
BTBTC06	**Cell Biology**	**3L - 0T - 2P**	**None**

COURSE OUTCOMES (COs):

1. The students will learn and understand that the cell is the basic structural and functional unit of life.
2. The students will be able to identify various organelles and analyze their importance in evolution and metabolism.
3. The students will be able to evaluate the effects of genetic changes during Cell growth and division.
4. The students will be apple to utilize the principles of signal transduction in evaluating the response of cell against any stimulus.
5. The students will be able to apply the basics concepts of cell biology in improving human health and life expectancy.

COURSE CONTENT:

Module I

Introduction to Cell Biology: Historical developments, microscopy, cell theory, cell size shape and architecture

Cellular Organelles: Biogenesis of cellular organelles, cell wall and their structural organization, nucleus, endoplasmic reticulum, golgi bodies, lysosomes, mitochondria, chloroplasts, microbodies - organization, structure and function.

Module II

Cell Division: Cell Cycle, Mitosis, Meiosis, Regulation of cell cycle

Membrane Transport: Plasma membrane – Structure and function, Principles of Membrane Transport, Carrier Proteins and Active Membrane Transport, Ion channels and Electrical Properties of Membranes, Ion pumps

Module III

Cell Signaling and Signal Transduction: General principles of cell signaling, receptors, types of receptors, cell to cell signalling: signaling via G-Protein-linked cell surface receptors, signaling via Enzyme-linked cell-surface receptors, ion and voltage gated channels, second messengers, Intracellular signaling

Module IV

Cytoskeleton and Cell motility: Cytoskeleton proteins, Microtubules, Intermediate Filaments and Microfilaments – structure, composition and function.

Module V

Apoptosis: Death domains, Direct Signal transduction, Caspases, mitochondrial regulation, apoptosome, Apoptosis in health and disease.

Cancer: Oncogenes, tumor suppressor genes, types of cancer, prevention and cure.

PRACTICAL:

1. Development of working knowledge of the Microscope and its Applications.
2. To observe distinguishing features of prokaryotic cell under microscope.
3. To observe distinguishing features of different eukaryotic cells under microscope.
4. To carry out vital staining of mitochondria, DNA and RNA in epithelial cells.
5. Preparation of blood smear and differential staining of blood cells.

6. To study different stages of Meiotic cell divisions.
7. To study different stages of Mitotic cell divisions.
8. To learn counting cells using hemocytometer.

SUGGESTED READINGS:

1. H. Lodish, A. Berk, S.L. Zipursky, P. Matsudaira, D. Baltimore and J. Darnell, "*Molecular Cell Biology*", W H Freeman and Company.
2. B. Alberts, D. Bray, A. Johnson, J. Lewis, M. Roff, K. Robert, P. Walter and K. Roberts, "*Essential Cell Biology: An Introduction to the Molecular Biology of the Cell*", Garland Publishers.
3. S.F. Gilbert, "*Developmental Biology*", Sinauer Associates Inc.
4. F.T. Longo, "*Fertilization*", Chapman and Hall.
5. L.P. Freedman, "*Molecular Biology of Steroid and Nuclear Hormone Receptors*", Birkhuser.
7. E.B. Wilson, "*Cell in Development and Inheritance*", Macmilan.
8. D.M Prescott, "*Reproduction in Eukaryotic Cells*", Academic press.
9. Ethan Bier, "*The Coiled Spring by. Publisher*", Cold Spring Harbor Press.
10. URL: http://www.cellbio.com

Digital Resources:

1) https://www.exploratorium.edu/education/teacher-institute/digital-teaching-boxes/cell-biology
2) http://www.scienceprofonline.com/instructors-corner/instructors-corner-vcbc.html
3) https://www.biozone.co.nz/iframes/iframe%20cell%20biology.html
4) https://www.cellsalive.com/
5) http://bcs.wiley.com/he-bcs/Books?action=index&itemId=1118206738&bcsId=7816

Course No.	Title of the Course	Course Structure	Pre-requisite
BTBTC07	Chemical Engineering Principles	3L - 1T - 0P	None

COURSE OUTCOMES (COs):

1. To understand the basic concepts of chemical engineering calculations
2. To apply the concepts of engineering calculations in calculating the material and energy requirements in production processes.
3. To analyze the heat and mass transfer limitation in chemical process and be able to choose the right equipment's.
4. To evaluate various reactor performance for homogenous reactions.
5. To design controllers for process control.

COURSE CONTENT:

MODULE I

Engineering Calculations – Variables, Dimensions, MODULEs and their conversions, Dimensional Homegeneity in Equations, Stoichiometry

MODULE II

Material and Energy Balances –General Mass balance equation,Different problems of material balance, Calculations involving MODULE processes and reactive systems, General Energy balance equation, enthalpy changes in chemical / biochemical reactions and in non-reactive processes, Energy balance calculations, Use of Steam tables, Heat of reaction and energy balance for microbial processes.

MODULE III

Chemical reaction engineering – Kinetics of homogenous reactions: Concepts of reaction rate, order of reaction and molecularity, Analysis of batch reactors for kinetic interpretation of data, Design equations for CSTR and plug flow reactors.

MODULE IV

Instrumentation and process control – Principles of measurement: error, accuracy and

sensitivity, Measurement of flow, pressure, temperature, level, pH, viscosity and chemical composition.
Basic concepts of feedback control, Controller hardware, Choice of controllers and settings. Introduction to advanced control systems

SUGGESTED READINGS:

1. D.M. Himmelblau, "*Basic Principles and Calculations in Chemical Engineering*".
2. E.I. Shaheen, "*Basic Principles of Chemical Engineering*".
3. G. Stephanopoulos, "*Chemical Process Control, An introduction to Theory and Practice*".
4. O. Levenspiel, "*Chemical Reaction Engineering*".
5. J.F. Richardson and D.G. Peacock, "*Coulson's and Richardson's Chemical Engineering*".
6. R.M. Felder and R.W. Rousseau, "*Elementary Principles of Chemical Processes*"
7. C.D Holland and R.G.Anthony, "*Fundamentals of Chemical Reaction Engineering*".
8. W.L. Luyben, "*Process Modelling, Simulation and Control for Chemical Engineers*".

Digital Resources:
https://ocw.mit.edu/courses/chemical-engineering/10-37-chemical-and-biological-reaction-engineering-spring-2007/lecture-notes/

Course No.	Title of the Course	Course Structure	Pre-requisite
BTBTC08	**Microbiology**	**3L - 0T - 2P**	**None**

COURSE OUTCOMES (COs):

1. Students will know about innovative approaches for working with microbes, will recognize microbes in various samples and will be able to define the microbialactivities in different environments.
2. Students will be able to compare and classify different category of microbes on the basis of their structure, nutrition and metabolism.
3. Students would have acquired the experimental skills to design suitable strategies toconduct experiments, collect data, analyze and interpret them.
4. Students will have an overall perspective of structural, functional and metabolic diversityof the microbes across diverse ecosystems that will enable them to design and createmethods, products and processes as necessary.
5. The students will be able to judge the validity of scientific statements or potential courses of action and detect errors or inconsistencies in such statements.
6. The students will thus be able to construct tables or figures to illustrate data or concepts and compose reports regarding abstract concepts.

COURSE CONTENT:

MODULE 1

Introduction and historical perspectives: Work of Antoni Van Leeuwenhoek, Louis Pasteur, Robert Kotch and others highlighting their approach and contribution to the science of microbiology.

MODULE 2

Methods in Microbiology: Theory andpractice of sterilization, Pure culture techniques, Enrichment culture techniques, methods of maintenance and preservation of pure cultures.

MODULE 3

Prokaryotic structure and function: Functional anatomy of bacteria: cell envelope, cell wall, cytoplasmic membrane, capsule, surface appendages, cytoplasm and cytoplasmic inclusions, spores.

Microbial diversity: Bacteria, Archaea, Eukarya: Algae, fungi, slime molds and protozoa. Viruses,

viriods and prions

MODULE 4

Microbial nutrition and growth: Principles of microbial nutrition and culture media, physiology of growth and factors affecting growth.

Microbial metabolism and Metabolic diversity: Basic considerations - role of ATP, reducing power and precursor metabolites; Fueling reactions in autotrophs and heterotrophs, ATP generation in phototrophs and chemotrophs (aerobic & anaerobic), Assimilation of nitrogen and sulfur, Strategy of biosynthesis.

MODULE 5

Microbes and the environment: Ecology of microbes, microbes as symbiotic partners, microbes and the earth, the evolution of microbes.

Microbial applications: in medicine, food, agriculture and energy

PRACTICAL:

1. Microbiology Laboratory Practices and Biosafety.
2. Preparation and sterilization of culture media for bacterial cultivation
3. Aseptic techniques: preparation of plates, slants and stabs, inoculation techniques
4. Study of Cell morphology by using different staining techniques
5. Study of microbial diversity in samples from soil, air and water
6. Isolation of pure culture and study of colony characteristics
8. Estimation of microbial growth and determination of growth rate
9. Factors affecting microbial growth

SUGGESTED READINGS:

1. M.T. Madigan, J.M. Martinko and J. Parker, "*Brock Biology of Microorganisms*", Prentice-Hall, Inc.
2. R.Y. Stanier, J.L. Ingraham, M.L. Wheelis and P.R. Painter "*General Microbiology*", Macmillan.
3. P.V. Vandenmark and B.L. Batzing, "*The Microbes – An Introduction to their Nature and Importance*", Benjamin Cummings.
4. Roger Y. Stanier, "*The Microbial World*", Prentice Hall.
5. H. G. Schlegel, "*General Microbiology*" Cambridge University Press 3. S.R. Maloy, J.E. Cronan and J.D. Freifelder, "*Microbial Genetics*", Bartlett Publishers.
6. J.G. Cappuccino and N. Sherman, "*Microbiology – A Laboratory Manual*", Addison-Wesley.
7. H.J. Benson, "*Microbiology Applications – A Laboratory Manual in General Microbiology*", Wm C Brown Publishers.
8. M.J. Pelczar, E.C.S. Chan and N.R. Kreig, "*Microbiology*", Tata McGraw Hill.
9. R.M. Atlas, "*Principles of Microbiology*", Wm C. Brown Publishers.

Digital Resources:

1. https://www.biozone.co.nz/biolinks/microbiology/
2. http://www.scienceprofonline.com/virtual-micro-main.html
3. https://www.tcd.ie/library/support/subjects/microbiology/index.php

4.3.4 SYLLABI OF PROGRAM CORE COURSES : IV SEMESTER

Course No.	Title of the Course	Course Structure	Pre-requisite
BTBTC09	Structural Biology	3L - 0T - 2P	None

COURSE OUTCOMES (COs):

After completion of course, the students will be able to:

1. Correlate the structure and physicochemical properties of the building blocks of biological

macromolecules, namely proteins and DNA.
2. Describe and discuss the various forces and thermodynamics involved in protein folding.
3. Apply the principles of structural biology to understand function and interactions of biological molecules.
4. Analyze, evaluate and design the techniques for stabilization of biomolecules in vitro.
5. Design the steps required for characterization of structure-function relationships.

COURSE CONTENT:

MODULE-I

Chemistry of amino acids and peptides (side chain structure and function in protein folding and functionality)**:** Secondary structure of proteins - helices, sheets, loops and turns, Structural and functional proteins. Domains and super-secondary structures; Tertiary structure of proteins, Quaternary structure and function, Sequence and structural motifs in proteins. Thermodynamics and molecular forces governing protein folding.

MODULE-II

Protein-ligand and protein-protein interactions: Lock and key versus handshake mechanism of substrate recognition, structural basis of recognition, reaction mechanisms of enzymes, G-Protein coupled receptors. Structure and function of an antibody, structure of hemoglobin, muscle proteins, forces governing protein-protein interactions.

MODULE-III

Protein solubility, protein stability and stabilization: salting in and salting out, Parameters affecting, enthalpic and entropic stabilization, mutations increasing stability, helix capping, Native, partially denatured and denatured proteins, Protein denaturation, Physical and chemical denaturants, Refolding.

MODULE-IV

DNA structure: covalent structure of DNA, base pairing, hydrogen bonding, DNA melting and annealing, difference between AT and GC pairing, DNA models, The Watson Crick model, Crystal structure of B-DNA, major and minor groves, base pair stacking, propellor twist, A and Z- DNA, triple stranded DNA.

MODULE-V

Special structures in DNA: Telomeric sequences and structure, G-quartets, palindromic and tandem repeats, Base pair flipping and DNA bulges, DNA methylation, Protein-DNA interactions, drug-DNA interaction.

PRACTICALS:

1. Chemical modification of proteins
2. Peptide mapping
3. Protein sequencing techniques
4. Analysis of CD data
5. Analysis and interpretation of Spectrophotometric data
6. Crystallization of Lysozyme
7. PDB structure and organization
8. Three-dimensional model building for DNA
9. Structural Classification of Proteins (SCOP) database
10. Graphical user interface for visualization of biomolecules

SUGGESTED READING:

1. R.H. Abeles, P.A. Frey and W.A. Jencks, "*Biochemistry*", Jones and Bartlett Publications

2. J. W. Mullin, "*Crystallization*", Butterworth Heinemann Publications

3. A. McPherson, "*Crystallization of Biological Macromolecules*", Cold Spring Harbor Laboratory Press.

4. D. Freifelder, "*Essentials of Molecular Biology*", Jones and Bartlett Publications.

5. B. Lewin, "*Genes VII*", Oxford University Press.

6. A.M. Lesk, "*Introduction to Protein Architecture: The Structural Biology of Proteins*", Oxford University Press.

7. C. Branden and J. Tooze, "*Introduction to Protein Structure*", Garland Publishing Company.

8. C.R. Cantor and P.R. Schimmel, "*Physical Chemistry*", W.H Freeman.

9. M. Perutz, "*Protein Structure*", Oxford University Press.

10. T.E. Creighton, "*Proteins (Structures and Molecular Properties)*", W.H. Freeman and Company.

DIGITAL RESOURCES:

1. https://www.ncbi.nlm.nih.gov/books/NBK21154/

Course No.	Title of the Course	Course Structure	Pre-requisite
BTBTC10	**Bioanalytical Techniques**	**3L - 1T - 0P**	**None**

COURSE OUTCOMES (COs):

After completion of course, the students will be able to:

1. Understand the concept of modern analytical techniques in qualitative and quantitative biochemical analyses
2. Explain experimental design and the application of statistical methods in the data analysis and interpretation of their significance
3. Apply the selection of an appropriate bioanalytical technique as per the experimental requirement and to justify the same
4. Appraise the bioanalytical techniques in multidisciplinary research as well as industrial operations
5. Critically evaluate the merits and limitations of any analytical techniques

COURSE CONTENT:

MODULE-I

Cell Disruption and Dehydration: Physical, Chemical, Mechanical and Enzymatic Methods, Lyophilization and osmotic dehydration

MODULE-II

Membrane-based techniques: Dialysis and Ultra-filtration

Centrifugation techniques: Principles of Centrifugation, High Speed and Ultra-centrifuges, and application of Density Gradient and Differential centrifugation, Speedvac.

MODULE-III

Electrophoresis: Principles of Electrophoresis, AGE, SDS-PAGE, Native PAGE, Isoelectric Focusing, 2-Dimensional electrophoresis, and their applications.

Chromatography: Principles of Chromatography - Ion exchange, Gel Filtration, Hydrophobic interaction, Affinity, GC, HPLC and FPLC and their applications

MODULE-IV

Spectroscopy: Principles of Spectroscopy-UV, Visible, Circular Dichroism, Infra-Red, FTIR, Raman, Mass Spectroscopy, Nuclear Magnetic Resonance and their applications.

Microscopy: Principles of Microscopy – Light and Electron Microscopy, Fluorescence and Confocal Microscopy and their applications

MODULE-V

Radioisotope techniques: Radio tracers, GM Counter, Proportional and Scintillation Counters, Autoradiography.

SUGGESTED READINGS:

1. J.F. Van Impe, "*Advanced Instrumentation, Data Interpretation, and Control of Biotechnological Processes*", Kluwer Academic.
2. K.E. Van Holde, "*Principles of Physical Biochemistry*", Prentice Hall.
3. Wilson and Walker's "*Principles and Techniques of Biochemistry and Molecular Biology*", Cambridge University Press
4. D.L. Pavia, G.M. Lampman and G. S. Kriz, "*Introduction to Spectroscopy*" Brooks Cole.
5. J. L. McHale, "*Molecular Spectroscopy*", Prentice Hall.
6. H. Günther, "*NMR Spectroscopy: Basic Principles, Concepts, and Applications in Chemistry*", John Wiley and Sons Ltd.
7. L. Que, "*Physical Methods in Bioinorganic Chemistry: Spectroscopy and Magnetism*", Junior University Science Books.
8.https://nptel.ac.in/courses/102/107/102107028/
9.https://www.asbmb.org/education/online-teaching

Course No.	Title of the Course	Course Structure	Pre-requisite
BTBTC11	**Molecular Biology**	**3L - 0T - 2P**	**None**

COURSE OUTCOMES (COs):

After course completion, students will be able to:

1. Develop a comprehensive understanding of the fundamental concepts in molecular biology.
2. Evaluate and judge the statements and information in molecular biology for their validity.
3. Acquire the ability and skills for analysis and interpretation of data and information pertaining to molecular biology.
4. Make use of their knowledge and skills in designing new experiments and developing innovative approaches for solving critical problems in molecular biology.
5. Demonstrate their understanding by explaining the concepts and will be able to recall and apply the facts learned earlier.

COURSE CONTENT:

MODULE-I

Introduction to Molecular Biology:Historical perspectives, Concept of gene and DNA as genetic material

Genes and Chromosome: Structure of DNA, Structure and organization of genes, chromosomes and chromatin

MODULE-II

DNA replication, DNA damage, repair and recombination

MODULE-III

Gene Expression-I (Transcription)**:** Mechanism of transcription in prokaryotic and eukaryotic systems, Regulation of Transcription; Modifications in RNA: 5'-cap formation, transcription termination, 3'-end processing and polyadenylation, Splicing, Editing, Nuclear export of mRNA and mRNA stability

MODULE-IV

Gene Expression-II (Translation)**:** Genetic code, Prokaryotic and Eukaryotic translation, Regulation of translation, Co-and post-translational modifications of proteins

MODULE-V

Gene regulation: The operon model (lac, ara, trp operons) in prokaryotes, regulation in Eukaryotes.

Regulatory RNAs: RNA mediated regulation of gene expression in prokaryotes and eukaryotes

PRACTICALS:

1. Preparation of solutions for Molecular Biology experiments
2. Determination of size of linear double stranded DNA fragment
3. Isolation of plasmid and genomic DNA from bacteria.
4. Agarose gel electrophoresis of plasmid and genomic DNAs
5. Isolation of total RNA from bacteria and its analysis by agarose gel electrophoresis
6. Quantification and purity determination of DNA /RNA using UV-Visible Spectroscopy
7. Native gel-electrophoresis for protein
8. Protein gel-electrophoresis under denaturing conditions

SUGGESTED READINGS:

1. B. Lewin, "*Gene IX*", Oxford University Press.
2. T.A. Brown, "*Genomes*", John Wiley and Sons Inc.
3. P.D. Dabre, "*Introduction to Practical Molecular Biology*", John Wiley and Sons Inc.
4. T.A. Brown, "*Molecular Biology LabFax*", Bios Scientific Ltd. Oxford.
5. B. Alberts, D. Bray, J. Lewis, M. Raff, K. Roberts and J.D. Watson, "*Molecular Biology of the Cell*", Garland Publishing.
6. J.D. Watson, A.M. Weiner and N.H. Hopkins, "*Molecular Biology of the Gene*", Addison-Wesley Publishing.
7. H. Lodish, A. Berk, S. Zipursky, P. Matsudaira, D. Baltimore and J.E Darnell, "*Molecular Cell Biology*", W.H. Freeman and Company.
8. J. Sambrook, E.F. Fritsch and T. Maniatis, "*Molecular Cloning: A Laboratory Manual (3-Volume Set)*", Cold spring Harbor Laboratory Press.
9. B. Alberts, D. Bray, A. Johnson, J. Lewis, M. Roff, K. Robert and P. Walter, "*Essential Cell Biology*" Garland Science, Taylor & Francis Group.
10. https://digital.wwnorton.com/48854/r/goto/cfi/54!/4

Course No.	Title of the Course	Course Structure	Pre-requisite
BTBTC12	**Genetics**	**3L - 0T - 2P**	**None**

COURSE OUTCOMES (COs):

After course completion, students will be able to:

1. To understand various mechanisms that underlie human inheritance
2. To analyze the role of genetic factors in health and disease and to be able to utilize the knowledge of the same for human welfare.
3. To apply the study in identification of patients with, or at risk of, a genetic condition and to communicate genetic information in an explicable way with an awareness of its impact on individual, family and society.
4. To gather and analyze current information about scientific and clinical applications of genetics.
5. To analyze, evaluate and apply different parameters, quantitatively governing population genetics.

COURSE CONTENT:

MODULE- I

Mendelian principles and Concept of gene: Dominance, segregation, independent assortment, Allele, multiple alleles, pseudoallele, complementation tests

Extensions of Mendelian principles: Codominance, incomplete dominance, gene interactions, pleiotropy, penetrance and expressivity, phenocopy, linkage and crossing over, sex linkage, sex-

limited, and sex influenced characters.

MODULE- II

Gene mapping methods: Linkage maps, tetrad analysis, mapping with molecular markers, mapping by using somatic cell hybrids, development of mapping population in plants.

Extra chromosomal inheritance: Inheritance of Mitochondrial and chloroplast genes, maternal inheritance.

MODULE-III

Microbial genetics: Methods of genetic transfers – transformation, conjugation, transduction.

Human genetics: Pedigree analysis, lod score for linkage testing, karyotypes, genetic disorders.

MODULE- IV

Quantitative genetics: Polygenic inheritance, heritability, and its measurements, QTL mapping.

Mutation: Types, causes, and detection, mutant types – lethal, conditional, biochemical, loss of function, a gain of function, germinal verses somatic mutants, insertional mutagenesis.

MODULE-V

Structural and numerical alterations of chromosomes: Deletion, duplication, inversion, translocation, ploidy, and their genetic implications.

Recombination: Homologous and non-homologous recombination including transposition.

PRACTICALS:

1. Mendelian deviations in dihybrid crosses
2. Karyotyping with the help of photographs
3. Pedigree charts of some common characters like blood group, color blindness & PTC testing.
4. To carry out mapping of genomic fragments.
5. To carry out population genetics study using sequence analysis.
6. To prepare metaphase plate of onion root tip cells.
7. Chi-Square Analysis.

SUGGESTED READINGS:

1. Peter J. Russell, "*iGenetics: A Molecular Approach*" Publisher Benjamin Cummings
2. D.P. Snustad and M.J. Simmons, "*Principles of Genetics*", John Wiley and Sons Inc.
3. M.W. Strickberger, "*Genetics*" Publisher", Prentice Hall College Division.
4. R. Levine, "*Genetics*", International Thomson Publishing.
5. U. Goodenough, "*Genetics*", International Thomson Publishing.
6. William S. Klug, Michael R. Cummings, & Charlotte A. Spencer."*Concepts of Genetics*" Benjamin Cummings
7. A.J.F. Griffiths, "*Introduction to Genetic Analysis*", W.H. Freeman and Company.
8. A.J. F. Griffiths, W.M. Gelbart, R.C. Lewontin and J.H. Miller, "*Modern Genetic Analysis: Integrating Genes and Genomes*", W.H. Freeman and Company.
9. https://ocw.mit.edu/courses/biology/7-03-genetics-fall-2004/
10. https://nptel.ac.in/courses/102/104/102104052/

Course No.	Title of the Course	Course Structure	Pre-requisite
BTBTC13	**Immunology**	**3L - 0T - 2P**	**None**

COURSE OUTCOMES (COs):

After completion of course, the students will be able to:

1. Differentiate between innate and adaptive immunity.
2. Understand the mechanisms and regulation of humoral and cell-mediated immune responses.
3. Apply principles of antigen-antibody interactions for clinical applications.

4. Evaluate the effects of hypersensitive reactions, autoimmunity and transplantation.
5. Apply the basics concepts of immunology for improvement in human health and life expectancy.

COURSE CONTENT:

MODULE-I

Introduction: Introduction to immunity, Innate and acquired immunity, clonal nature of immune response.

Cells and organs of the immune system: hematopoiesis and differentiation, B-Lymphocytes, T-Lymphocytes, monocytes, macrophages, dendritic cells, natural killer cells, neutrophils, eosinophils, basophils, mast cells, primary and secondary lymphoid organs

MODULE-II

Antigens: Immunogen, Epitope, Hapten, Adjuvants

Antibody: Structure, Function, BCR, Generation of antibody diversity, Hybridoma technology - Fusion of myeloma cells with lymphocytes, production of monoclonal antibodies and their applications

MODULE-III

Major Histocompatibilty Complex and Antigen Presentation: Structure and function of MHC molecules, Self MHC restriction, Exogenous and Endogenous pathways of antigen processing and presentation, T cell receptor, T & B cell maturation, activation and differentiation

MODULE-IV

Immune Effector Mechanisms: Cytokines, Complement system, Cell and Antibody mediated immunity: mechanism of T cell and NK cell mediated lysis, macrophage mediated cytotoxicity, antibody dependent cell mediated cytotoxicity

MODULE-V

Hypersensitivity, Autoimmunity, Graft vs Host Reaction, Vaccines

PRACTICALS:

1. To study different organs of immune system.
2. To determine the blood group type and Rh factor of a blood sample.
3. To study the immunodiffusion technique by Single Radial Immunodiffusion.
4. To study the reaction pattern of an antigen with a set of antibodies by Ouchterlony Double Diffusion method.
5. To learn the technique of Immunoelectrophoresis.
6. To detect the presence of *Salmonella* genus which causes enteric or Typhoid Fever by using qualitative slide agglutination test.
7. To learn the technique of Dot ELISA for the detection of an antigen.
8. To determine the antigen concentration by Antigen Capture ELISA method.

SUGGESTED READINGS:

1. Kuby, RA Goldsby, Thomas J. Kindt, Barbara, A. Osborne, "*Immunology, 6th Edition*", Freeman.
2. Janeway et al., "*Immunobiology, 4th Edition*", Current Biology Publications.
3. Paul, "*Fundamental of Immunology, 4th edition*", Lippencott Raven
4. https://www.immunology.org/public-information/immunology-related-activities-and-resources
5. https://www.aai.org/Education/Teaching-Resources

4.3.5 SYLLABI OF PROGRAM CORE COURSES : V SEMESTER

4.3.6 SYLLABI OF PROGRAM CORE COURSES : VI SEMESTER

4.4 DEPARTMENT ELECTIVE COURSES

4.4.1 LIST OF DEPARTMENT ELECTIVES

4.4.2 SYLLABI OF DEPARTMENT ELECTIVES COURSES : V SEMESTER

4.4.3 SYLLABI OF DEPARTMENT ELECTIVES COURSES : VI SEMESTER

4.4.4 SYLLABI OF DEPARTMENT ELECTIVES COURSES : VII & VIII SEMESTERS

4.5 OPEN ELECTIVE COURSES

4.5.1 LIST OF OPEN ELECTIVES

4.5.2 SYLLABI OF OPEN ELECTIVE COURSES : SCIENCES

4.5.3 SYLLABI OF OPEN ELECTIVE COURSES : HUMANITIES & SOCIAL SCIENCES

4.5.4 SYLLABI OF OPEN ELECTIVE COURSES : MANAGEMENT